Electrostatic Potential Energy

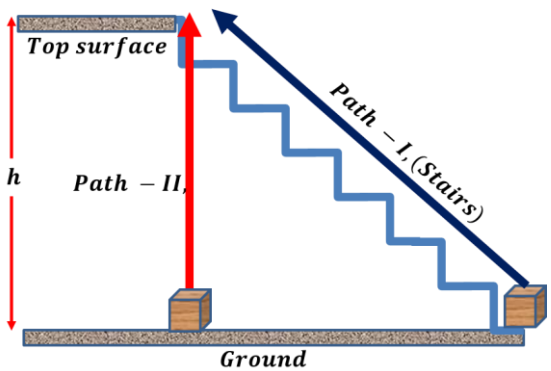
Part - 01

Electrostatic Potential Energy (EPE)

- ❖ EPE of a point charge, is the amount of work done in bringing the charge from infinity to the present position without changing its kinetic energy.
- ❖ EPE of a system of charges, is the minimum amount of work done in assembling the given charge system.
- ❖ An object may have EPE by virtue of
  (i) Its own charge called self-energy and
  (ii) Its relative position in the system of charges, called interaction energy.
  Therefore, total EPE = self-energy + interaction energy
- ❖ EPE is defined only for conservative field like gravitational field, electrostatic field etc.

Properties of conservative field:

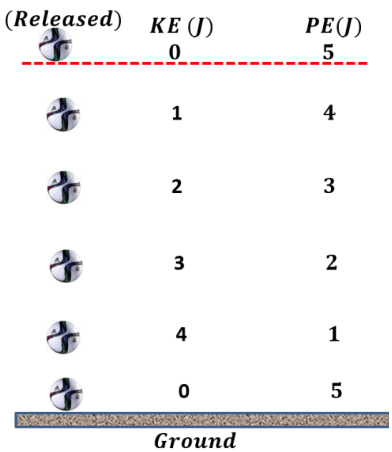
(i) Work done is independent of path followed.



In both cases work done against gravity will be same $\Rightarrow w_I = w_{II}$

(ii) Work done against conservative field/force, gets stored in the form of potential energy.

(iii) Sum of KE and PE at any instant remains constant [Conservation of Mechanical Energy (COME)].



From Work Energy Theorem

$W_{\text{external agent}} + W_{\text{internal conservative field}} + W_{\text{internal non-conservative field}} = \Delta KE$

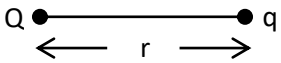
$W_{\text{internal non-conservative field}} = 0$ (Assumed)

$W_{\text{external agent}} + W_{\text{internal conservative field}} = \Delta KE$

For $\Delta KE = 0$; $W_{\text{external agent}} = -W_{\text{internal conservative field}} = U_f - U_i$ ($\because W_{\text{internal conservative field}} = U_i - U_f$)

$W_{\text{external agent}} = U_f - U_i$

Electrostatic Potential Energy (EPE) of two point charge system:



To find this, consider Q is at rest and q is brought from ∞ to r distance from the rest charge. (Here reference is at infinity)

At any instant suppose charge **q** is at distance **x** from **Q.**

At this instant electric force on q is



$F_{elec} = \dfrac{kQq}{x^2}$ (rightwards)

Then, $\Delta U = (W_{\infty \to r})_{ext.} = \int_{\infty}^{r} \vec{F}_{ext}.d\vec{r}$

$\Rightarrow U_r - U_\infty = \Delta U = -(W_{\infty \to r})_{elec} = -\int_{\infty}^{r} \vec{F}_{elec} d\vec{x}$

$\Rightarrow U_r - U_\infty = -\int_{\infty}^{r} \vec{F}_{elec} d\vec{x} \qquad \left[\because \vec{F}_{elec} \parallel d\vec{x}\right]$

$\Rightarrow U_r - U_\infty = -\int_{\infty}^{r} \dfrac{kQq}{x^2} dr = kQq\left[+\dfrac{1}{x}\right]_{\infty}^{r}$

$\Rightarrow U_r - U_\infty = kQq\left[\dfrac{1}{r} - \dfrac{1}{\infty}\right] = \dfrac{kQq}{r}$

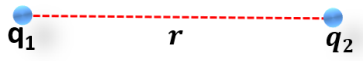
As we consider infinity as reference & $U_\infty = 0$, so we have

U-0 = $\dfrac{kQq}{r}$

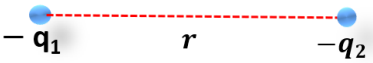
$\Rightarrow U = \dfrac{kQq}{r}$

Note : *Always put sign of the charge in this formula.*

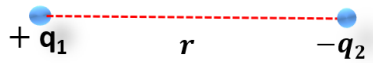
- Potential energy under repulsive forces is positive.
- Potential energy under attractive forces is negative.

(1) 

$U = \frac{kq_1q_2}{r}$

(2) 

$U = \frac{kq_1q_2}{r}$

(3) 

$U = -\frac{kq_1q_2}{r}$

❖ **Amount of work done in changing the configuration of charge particles.**

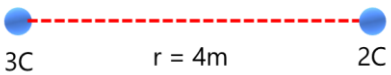
$W = \Delta U = U_f - U_i$

Where U_f = EPE of final configuration of charge particles

U_i = EPE of initial configuration of charge particles

Illustration 1 :

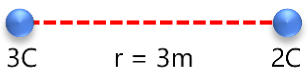
Find potential energy of the system shown. Also find work done in decreasing the distance by 1m.



Solution:

$$U_i = \frac{kq_1q_2}{r} = \frac{9\times10^9(3)(2)}{4} = 13.5\times10^9\ \text{J}$$

When distance decreases by 1m, separation between charges becomes 3m



$$\because U_f = \frac{9\times10^9(3)(2)}{3} = 18\times10^9\ \text{J}$$

Work done in this process

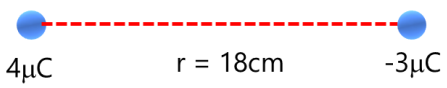
$W = U_f - U_i = 18 \times 10^9 - 13.5\times 10^9 \Rightarrow W = 4.5 \times 10^9\ \text{J}$

Illustration 2 :

For the given charge system, find –

(1) PE of system

(2) Work done in carrying a charge to infinity



Solution:

(1) $U = \frac{kq_1q_2}{r} = \frac{(9\times10^9)(4\times10^{-6})(-3\times10^{-6})}{(18\times10^{-2})}$

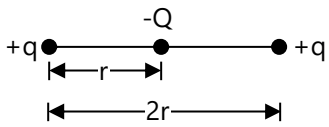
$U = -0.6J$

(2) $W = U_f - U_i$

$W = 0 - (-0.6J) = +0.6J$

Illustration 3:

Figure shows an arrangement of three-point charges. The total potential energy of this arrangement is zero. Calculate the ratio $\frac{q}{Q}$.



Solution:

$\because U_{sys} = 0$

$\Rightarrow \frac{1}{4\pi\varepsilon_0}\left[\frac{-qQ}{r}+\frac{(+q)(+q)}{2r}+\frac{Q(-q)}{r}\right] = 0$

$\Rightarrow -Q + \frac{q}{2} - Q = 0$

$\Rightarrow 2Q = \frac{q}{2} \Rightarrow \frac{q}{Q} = \frac{4}{1}$

Electrostatic Potential Energy of a System of Charges

Part - 02

Electrostatic Potential Energy of a System of Charges

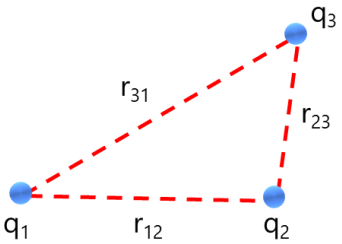
It is defined as work done by external agent or work done against electric field in bringing charges from infinity to their respective positions, in forming system of charges.

Example –

(i) For 3-point charge system -

$U_S = U_{12} + U_{23} + U_{31}$

$$U_s = \frac{kq_1q_2}{r_{12}} + \frac{kq_2q_3}{r_{23}} + \frac{kq_3q_1}{r_{31}}$$



(ii) For 4-point charge system –

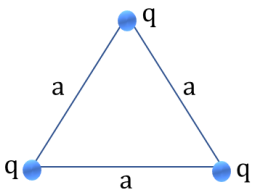
$U_S = U_{12} + U_{13} + U_{14} + U_{23} + U_{24} + U_{34}$

Similarly, for n point charge system

No. of pairs $= \frac{n(n-1)}{2}$

Illustration 1:

Consider the charge system shown, find
(i) Potential energy of any one of the charges.
(ii) Potential energy of the system.
(iii) Work done in associating the system.



Solution:

(i) $U_1 = U_{12} + U_{13} = \frac{kq^2}{a} + \frac{kq^2}{a} = \frac{2kq^2}{a}$

(ii) $U_S = U_{12} + U_{13} + U_{23} = \frac{kq^2}{a} + \frac{kq^2}{a} + \frac{kq^2}{a} = \frac{3kq^2}{a}$

(iii) $W = U_f - U_i = U_s - U_\infty = \frac{3kq^2}{a} - 0 = \frac{3kq^2}{a}$

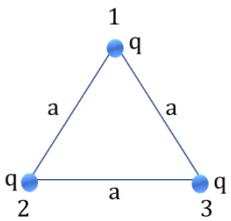
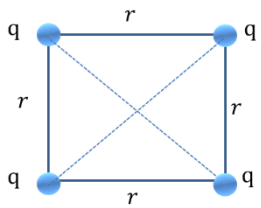


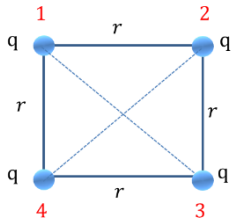
Illustration 2 :

Consider the point charge system shown, find
(i) Potential energy of any one of the charges.
(ii) Potential energy of the system.
(iii) Work done in dissociating the system.



Solution:

(i) $U_1 = U_{12} + U_{13} + U_{14} = \frac{2kq^2}{r} + \frac{kq^2}{r\sqrt{2}} = \frac{kq^2}{r}\left(2 + \frac{1}{\sqrt{2}}\right)$



(ii) $U_s = U_{12} + U_{23} + U_{34} + U_{41} + U_{13} + U_{42}$

$$U_S = \underbrace{\frac{kq_1q_2}{r} + \frac{kq_2q_3}{r} + \frac{kq_3q_4}{r} + \frac{kq_4q_1}{r}} + \underbrace{\left\{\frac{kq_1q_3}{\sqrt{2}r} + \frac{kq_4q_2}{\sqrt{2}r}\right\}}$$

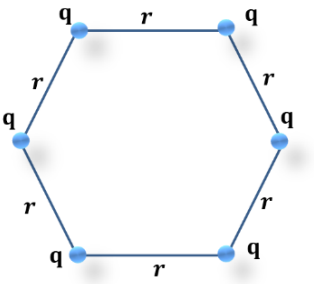
$\because$ If all charges are identical,

$$U_S = \frac{4kq^2}{r} + \frac{2kq^2}{\sqrt{2}r} \qquad U_S = \frac{kq^2}{r}(4 + \sqrt{2})$$

(iii) $W = U_f - U_i = U_\infty - U_s = -\frac{kq^2}{r}\left(\sqrt{2} + 4\right)$

Illustration 3:

Consider the point charge system shown, find
(i) Potential energy of any one of the charges.
(ii) Potential energy of the system.
(iii) Work done in dissociating the system.

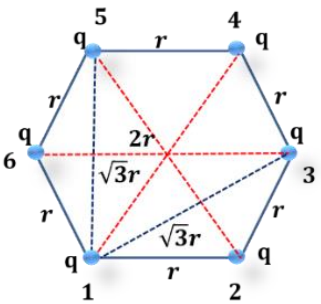


Solution:

(ii) $U_1 = U_{12} + U_{13} + U_{14} + U_{15} + U_{16}$

$$U_1 = \frac{kq^2}{r} + \frac{kq^2}{\sqrt{3}r} + \frac{kq^2}{2r} + \frac{kq^2}{\sqrt{3}r} + \frac{kq^2}{r}$$

$$U_1 = \left(\frac{2kq^2}{r} + \frac{2kq^2}{\sqrt{3}r} + \frac{kq^2}{2r}\right)$$



(ii) $U_s = \frac{1}{2}(U_1 + U_2 + U_3 + U_4 + U_5 + U_6)$

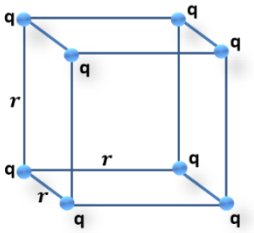
$U_s = \frac{1}{2}(6U_1) = 3U_1$ [$U_1 = U_2 = U_3 = U_4 = U_5 = U_6$, since identical charges and symmetric distribution]

$$U_s = 3U_1 = 3\left(\frac{2kq^2}{r} + \frac{2kq^2}{\sqrt{3}r} + \frac{kq^2}{2r}\right)$$

(iii) $W = U_f - U_i = U_\infty - U_s = -3\left(\frac{2kq^2}{r} + \frac{2kq^2}{\sqrt{3}r} + \frac{kq^2}{2r}\right)$

Illustration 4:

Consider the point charge system shown, find
(i) Potential energy of any one of the charges.
(ii) Potential energy of the system.
(iii) Work done in dissociating the system.



Solution:

(i) $U_1 = U_{12} + U_{13} + U_{14} + U_{15} + U_{16} + U_{17} + U_{18}$

$$U_1 = \left(\frac{3kq^2}{r} + \frac{3kq^2}{\sqrt{2}r} + \frac{kq^2}{\sqrt{3}r} \right)$$

(ii) $U_s = \frac{1}{2} = (8U_1) = 4U_1$ [Identical charges and symmetric distribution]

$$U_s = 4\left(\frac{3kq^2}{r} + \frac{3kq^2}{\sqrt{2}r} + \frac{kq^2}{\sqrt{3}r} \right)$$

(iii) Work done = $U_f - U_i = U_\infty - U_s = -U_s = -4\left(\frac{3kq^2}{r} + \frac{3kq^2}{\sqrt{2}r} + \frac{kq^2}{\sqrt{3}r} \right)$

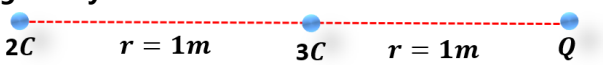
Note : In case of cube -

Variety	distance	Number of pairs	$N_p = \frac{8\times 7}{2} = 28$
Side	$\sqrt{1}a$	12	
Face diagonal	$\sqrt{2}a$	12	
Main diagonal	$\sqrt{3}a$	4	

$$U_{system} = \frac{12kq^2}{a} + \frac{12kq^2}{\sqrt{2}a} + \frac{4kq^2}{\sqrt{3}a}$$

Illustration 5 :

For what value of Q, EPE of the given system is zero.



Solution:

$$U_s = \left(\frac{k(2\times 3)}{1} + \frac{k(3\times Q)}{1} + \frac{k(2\times Q)}{2} \right)$$

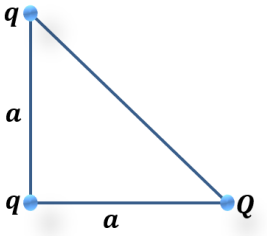
According to the given condition, $U_s = 0$

$$\left(\frac{k(2\times 3)}{1} + \frac{k(3\times Q)}{1} + \frac{k(2\times Q)}{2} \right) = 0$$

$6 + 3Q + Q = 0 \Rightarrow 4Q = -6 \Rightarrow Q = -1.5C$

Illustration 6 :

For what value of Q, EPE of the system shown is zero.



Solution:

$$U_s = \left(\frac{k(q \times q)}{a} + \frac{k(q \times Q)}{a} + \frac{k(q \times Q)}{\sqrt{2}a} \right)$$

According to the given condition

$$\left(\frac{k(q \times q)}{a} + \frac{k(q \times Q)}{a} + \frac{k(q \times Q)}{\sqrt{2}a} \right) = 0$$

$$\Rightarrow \left(q^2 + (q \times Q) + \frac{(q \times Q)}{\sqrt{2}} \right) = 0$$

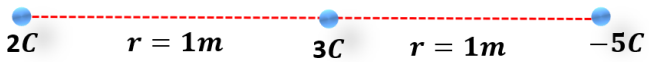
$$\Rightarrow \left(q + (Q) + \frac{(Q)}{\sqrt{2}} \right) = 0$$

$$\Rightarrow (Q) + \frac{(Q)}{\sqrt{2}} = -q$$

$$\Rightarrow \quad Q = \frac{-q\sqrt{2}}{1+\sqrt{2}}$$

Illustration 7 :

Find work done in shifting -5C charge to a point 1m equidistance from both the remaining charges.



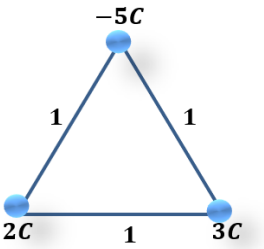
Solution:

$$W = \Delta U = (U_f) - (U_i)$$

$$\Delta U = \left[\frac{k(2)(-5)}{1} + \frac{k(2) \times (3)}{1} + \frac{k(3) \times (-5)}{1} \right] - \left[\frac{k(2) \times (3)}{1} + \frac{k(3) \times (-5)}{1} + \frac{k(2) \times (-5)}{2} \right]$$

= (–10k + 6k – 15k) – (6k – 15k – 5k)

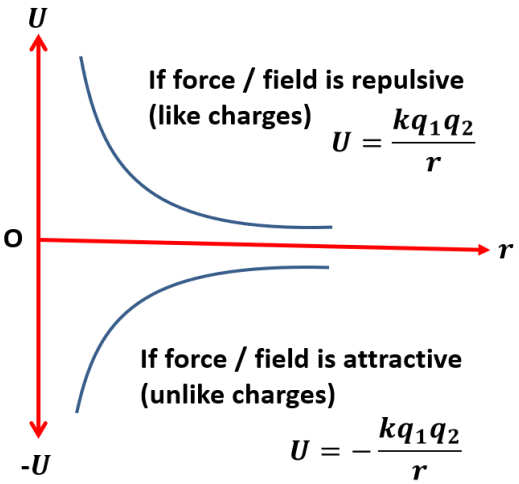
= -5k = – 45×10^{9}J



Problems Based on Potential Energy

Part - 03

Variation of U with r for two-point charge system



For like charge system



According to $U = \frac{kq_1q_2}{r}$, As $r \uparrow, U \downarrow$ (vice versa)

For unlike charge system



According to $U = -\frac{kq_1q_2}{r}$, As $r \uparrow, U \uparrow$ (vice versa)

Illustration 1 :

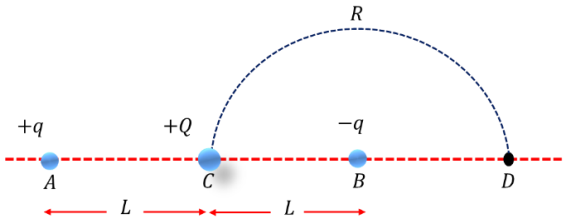
If separation between two charges is increased, whether EPE of the system increases or decreases.

Solution:

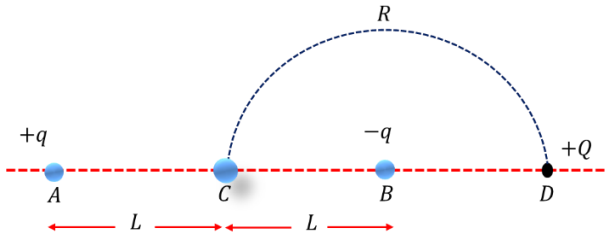
$\therefore$ It may increase or decrease depending upon sign of charges of system.

Illustration 2 :

Find work done in transferring the charge +Q via semicircular path CRD centered at B.



Solution:

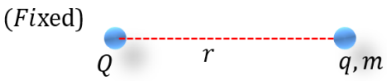


$$W = \Delta U = (U_f) - (U_i)$$

$$W = \left[\frac{k(q)(-q)}{2L} + \frac{k(-q)(Q)}{L} + \frac{k(q)(Q)}{3L}\right] - \left[\frac{k(q)(Q)}{L} + \frac{k(Q)(-q)}{L} + \frac{k(q)(-q)}{2L}\right] = -\frac{2kqQ}{3L}$$

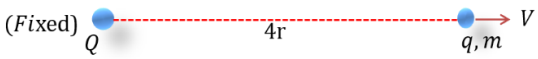
Illustration 3 :

Charge (q, m) is released from rest position as shown in figure. Find the speed of the particle when its separation from charge Q becomes 4r.



Solution:

On releasing q, let its speed becomes v at separation of 4r then,



Using COME, $K_i + U_i = K_f + U_f$

$$\Rightarrow 0 + \frac{k(Q)(q)}{r} = \frac{1}{2}mv^2 + \frac{k(Q)(q)}{4r}$$

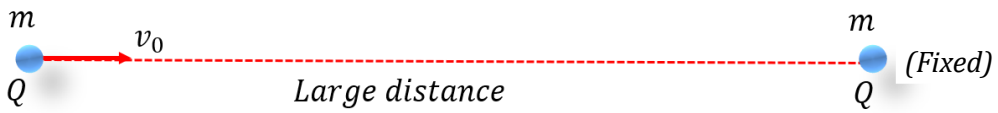
$$\Rightarrow \frac{1}{2}mv^2 = \frac{kQq}{r} - \frac{kQq}{4r}$$

$$\Rightarrow \frac{1}{2}mv^2 = \frac{3kQ}{4r}$$

$$\Rightarrow v = \sqrt{\frac{3kQq}{2mr}}$$

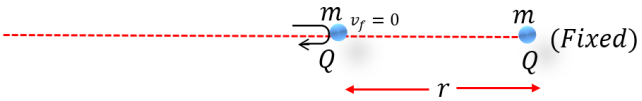
Illustration 4 :

Find minimum distance of approach between two charge particles, shown below.



Solution:

At minimum distance of approach particle will stop for a moment and then return back



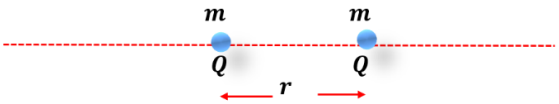
Using COME, $K_i + U_i = K_f + U_f$

$$\Rightarrow \frac{1}{2}mv_0^2 + \frac{kQ^2}{\infty} = 0 + \frac{kQ^2}{r}$$

$$\Rightarrow r = \frac{2kQ^2}{mv_0^2}$$

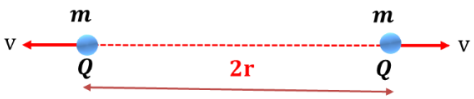
Illustration 5 :

Two identical charge particles are released from initial separation r (as shown), find velocities of each charge particle when their separation becomes (i) 2r and (ii) infinity



Solution:

(i) When separation becomes 2r, see figure



From momentum conservation both will have equal speeds then

$K_i + U_i = K_f + U_f$

$$\Rightarrow 0 + \frac{kQ^2}{r} = 2 \times \frac{1}{2}mv^2 + \frac{kQ^2}{2r}$$

$$\Rightarrow 0 + \frac{kQ^2}{2r} = mv^2$$

$$\Rightarrow v = \sqrt{\frac{kQ^2}{2mr}}$$

(ii) When the separation becomes infinity, EPE would be zero.

Using COME, $K_i + U_i = K_f + U_f$

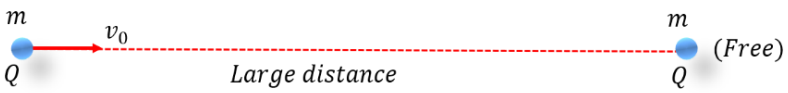
$$\Rightarrow 0+\frac{kQ^2}{r}=2\times\frac{1}{2}mv^2+0$$

$$\Rightarrow \frac{kQ^2}{r}=mv^2$$

$$\Rightarrow v=\sqrt{\frac{kQ^2}{mr}}$$

Illustration 6 :

Find distance of closest approach for the given system of two charge particles, shown below.

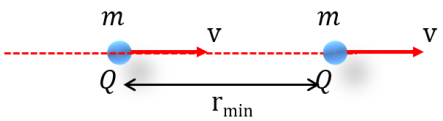


Solution:

Initial situation is -



Finally, at minimum distance both will acquire same speed (v) –



As there is no external force acting on system, so we use Conservation of Linear Momentum (COLM) to find final velocities of charge particles

$\Rightarrow mv_0 = mv + mv$

$$\Rightarrow v=\frac{v_0}{2}$$

Now we use COME to find minimum distance of approach

$\Rightarrow K_i + U_i = K_f + U_f$

$$\Rightarrow \frac{1}{2}mv_0^2+0=\frac{1}{2}mv^2+\frac{1}{2}mv^2+\frac{kQ^2}{r}$$

$$\Rightarrow \frac{1}{2}mv_0^2=2\left[\frac{1}{2}m\left(\frac{v_0}{2}\right)^2\right]+\frac{kQ^2}{r}$$

$$\Rightarrow \frac{1}{2}mv_0^2=m\left(\frac{v_0^2}{4}\right)+\frac{kQ^2}{r}$$

$$\Rightarrow \frac{1}{4}mv_0^2=\frac{kQ^2}{r}$$

$$\Rightarrow r_{min}=\frac{4kQ^2}{mv_0^2}$$

Illustration 7 :

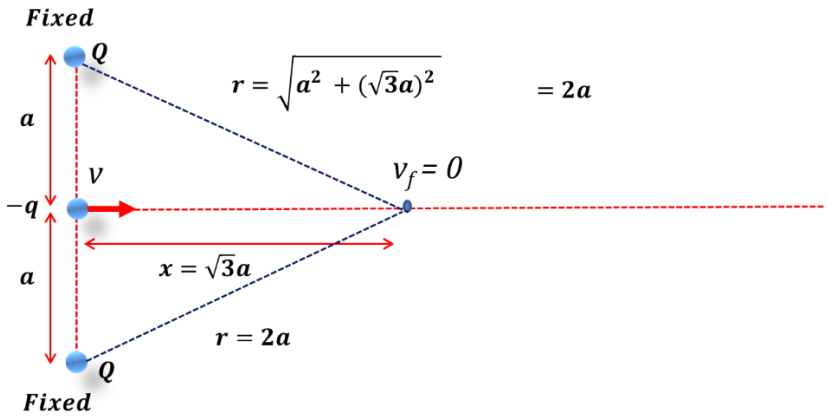
Two identical charge particles (each +Q) are fixed at positions (0,a) and (0,-a). A negative charge –q, m placed at origin, is projected along positive x axis. Find the velocity of projection so that it comes to rest (i) at $x=\sqrt{3}a$ and (ii) at infinity.

Solution:

Let velocity of projection of -q is v

(i) When -q comes to rest at $x=\sqrt{3}a$ then using COME -

$K_i + U_i = K_f + U_f$



$$\Rightarrow \frac{1}{2}mv^2+\frac{kQ^2}{2a}+\left(-2\frac{kQq}{a}\right)=0+\left(-2\frac{kQq}{2a}\right)+\frac{kQ^2}{2a}$$

$$\Rightarrow \frac{1}{2}mv^2-\frac{2kQq}{a}=-\frac{kQq}{a}$$

$$\Rightarrow \frac{1}{2}mv^2=\frac{kQq}{a}$$

$$\Rightarrow v=\sqrt{\frac{2kQq}{ma}}$$

(ii) When charge goes to infinity, then again using COME

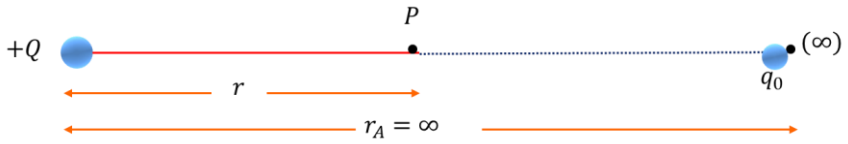
$K_i + U_i = K_f + U_f$

$$\Rightarrow \frac{1}{2}mv^2+\frac{kQ^2}{2a}+\left(-2\frac{kQq}{a}\right)=0+\frac{kQ^2}{2a}$$

$$\Rightarrow v=\sqrt{\frac{4kQq}{ma}}$$

Electric Potential

Electric potential at a point in electric field is defined as amount of work done by external agent in bringing a unit test positive charge from infinity to that point against the electric field.



$$V_P = \frac{W_{\infty \to P}}{q_0}$$

- ❖ It is scalar quantity
- ❖ S.I. Unit of potential is volt = $\frac{joule}{coulomb}$ and its dimensional formula is $[M^1L^2T^{-3}I^{-1}]$.
- ❖ Potential can be positive, negative and even zero. Hence charges must be taken with sign.
- ❖ Electric potential always decreases in the direction of electric field

$$V_P = \left(\frac{W_{agent}}{q_0}\right)_{\Delta K=0} \qquad V_P = \left(-\frac{W_{electric}}{q_0}\right)_{\Delta K=0}$$

$$W_{ext} = \Delta U = (U_f - U_i) \qquad W_{ele} = -\Delta U = (U_i - U_f)$$

Illustration 8 :

A point charge of 20μC is brought from infinity to a point in an electric field. If -40 μJ of work had to be done against the field then find potential at that point.

Solution:

$$V_P = \left(\frac{W_{agent}}{q_0}\right)_{\Delta K=0}$$

$$V_P = \left(\frac{-(40\times10^{-6})}{(20\times10^{-6})}\right)_{\Delta K=0}$$

So electric potential at that point is -2V.

Illustration 9 :

When a point charge of 20μC is brought from infinity to a point in an electric field, -20μJ of work is done by the electric field. Then what amount of work would have to be done by the external agent, if a 40μC charge brought from infinity to the same point ?

Solution:

$$V_P = -\left(\frac{W_{(\infty\to P)ele}}{q_0}\right) \quad \Rightarrow \quad V_P = -\left(\frac{-(20\times10^{-6})}{(20\times10^{-6})}\right)$$

$V_P = 1V$

Now from

$$V_P = \left(\frac{W_{(\infty\to P)}ext}{q}\right) \Rightarrow W_{(\infty\to P)}ext = qV_P = (40\times10^{-6})\times1 \quad \Rightarrow \quad W_{(\infty\to P)ext} = 40\mu J$$

Illustration 10 :

A charge particle of 2C is released from a point where electric potential is 10V. Find kinetic energy of the particle when charge particle reaches to infinity.

Solution:

Initially



$KE_i = 0$

$U_i = 20J$

At infinity $KE_f = ?, \quad U_f = 0$

Using COME $KE_i + U_i = KE_f + U_f$

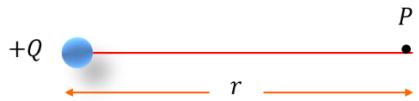
$0 + 20 = KE_f + 0 \Rightarrow KE_f = 20J$

Electrostatic Potential Due to a Point Charge

Part - 04

Electrostatic Potential due to a point charge:

Electric potential at a point in electric field is defined as amount of work done in bringing a unit positive charge from infinity to given point against the electric field. (without change in KE)



If a test charge q_0 is brought from ∞ to point P then

Work done in this process $W_{\infty \to P} = \frac{kQq_0}{r}$

According to definition, $V_P = \frac{W_{\infty \to P}}{q_0} = \frac{kQq_0}{q_0 r} = \frac{kQ}{r} \Rightarrow \boxed{V_P = \frac{kQ}{r}}$

Conclusion :

- Potential due to a positive point charge



➢ Potential due to a negative point charge

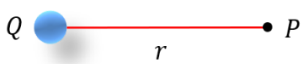


Where *reference potential is 0 at infinity.*

Illustration 1 :

Calculate potential due to a charge 5×10^{-7} C at a point P located 9 cm away. Also find work done in bringing an another charge of 3×10^{-8}C from infinity to the point.

Solution:



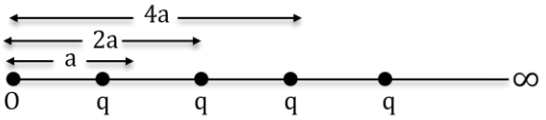
$$V_p = \frac{kQ}{r} = \frac{9\times10^9 \times 5\times10^{-7}}{9\times10^{-2}} = 5\times10^4 \text{ V}$$

$W = q(\Delta V) = q\,(V_p - V_\infty) = 3\times10^{-8}\,(5\times10^4 - 0) = 15\times10^{-4}$ J = 1.5 mJ

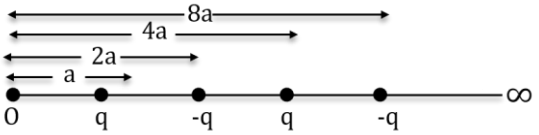
Illustration 2 :

Find potential at origin O due to the given charge distributions.

(i)



(ii)



Solution:

(i) $$V_0 = \frac{kq}{a} + \frac{kq}{2a} + \frac{kq}{4a} + \frac{kq}{8a} + ...\infty$$

$$= \frac{kq}{a}\left[1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + ...\infty\right]$$

$$= \frac{kq}{a}\left[\frac{1}{1-\frac{1}{2}}\right]$$

$$\Rightarrow V_0 = \frac{2kq}{a}$$

(ii) $$V_0 = \frac{kq}{a} - \frac{kq}{2a} + \frac{kq}{4a} - \frac{kq}{8} + ...\infty$$

$$= \frac{kq}{a}\left[1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + ...\infty\right]$$

$$= \frac{kq}{a}\left[\frac{1}{1+\frac{1}{2}}\right]$$

$$\Rightarrow V_0 = \frac{2kq}{3a}$$

Electrostatic Potential Due to a System of Charges

Part - 05

Other physical interpretation of electric potential:

In electrostatic field, the electric potential (due to some source charges) at a point P is defined as the work done by external agent in taking a unit point positive charge from a reference point (generally taken at infinity) to that point P without changing its kinetic energy.

Mathematical representation:

If $(W_{\infty \to P})_{ext}$ is the work required in moving a point charge q from infinity to a point P, the electric potential of the point P is

$$V_p = \left.\frac{(W_{\infty \to p})_{ext}}{q}\right]_{\Delta K=0} = \left.-\frac{(W_{\infty \to p})_{elec}}{q}\right]_{\Delta K=0}$$

Note:

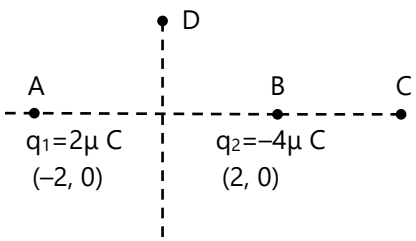
- $(W_{\infty \to P})_{ext}$ can also be called as the work done by external agent against the electric force on a unit positive charge due to the source charge.
- Write both W and q with proper sign.

Properties :

- Electric potential at a point is also equal to the negative of the work done by the electric field in taking the point charge from reference point (i.e. infinity) to that point.
- Electric potential due to a positive charge is always positive and due to negative charge. It is always negative except at infinity. (Taking $V_\infty = 0$).
- $V = V_1 + V_2 + V_3 + ...$

Illustration 1:

Two-point charges $2\mu C$ and $-4\mu C$ are situated at points $(-2m, 0m)$ and $(2\,m, 0\,m)$ respectively. Find out potential at point $C(4\,m, 0\,m)$ and $D(0\,m, \sqrt{5}\,m)$.



Solution:

Potential at point C

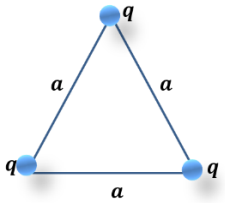
$$V_C = V_{q_1} + V_{q_2} = \frac{k(2\mu C)}{6} + \frac{k(-4\mu C)}{2} = \frac{9\times10^9\times2\times10^{-6}}{6} - \frac{9\times10^9\times4\times10^{-6}}{2} = -15000\ V.$$

Similarly, $V_D = V_{q_1} + V_{q_2} = \frac{k(2\mu C)}{\sqrt{(\sqrt{5})^2+2^2}} + \frac{k(-4\mu C)}{\sqrt{(\sqrt{5})^2+2^2}} = \frac{k(2\mu C)}{3} + \frac{k(-4\mu C)}{3} = -6000V.$

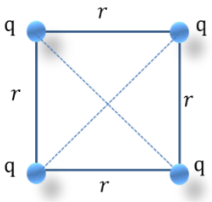
Illustration 2 :

Find potential at the centre of the charge systems shown-

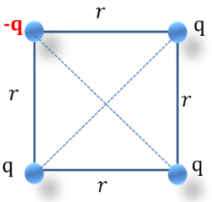
(i)



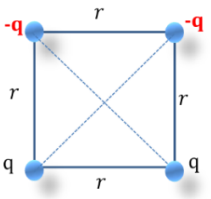
(ii)



(iii)

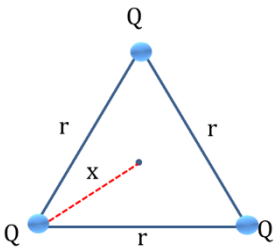


(iv)



Solution:

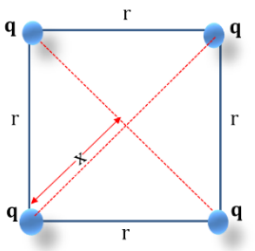
(i) Here value of x is $x = \frac{r}{\sqrt{3}}$



Potential at centre due to all three charges is

$$V = \frac{3kQ}{x} = \frac{3\sqrt{3}kQ}{r}$$

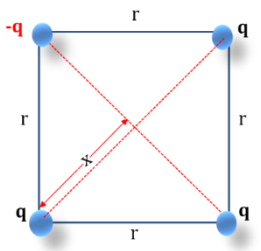
(ii) Here value of $x = \frac{r}{\sqrt{2}}$



Potential at centre due to all four charges is

$$V = \frac{4kq}{x}$$

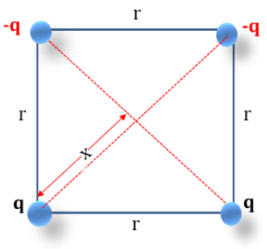
$$V = \frac{4\sqrt{2}kq}{r}$$



(iii) Potential at centre due to all four charges is

$$V = \frac{3kq}{x} - \frac{kq}{x} = \frac{2kq}{x} = \frac{2\sqrt{2}kq}{r}$$

(iv) Potential at centre due to all four charges is



$$V = \frac{2kq}{x} - \frac{2kq}{x} = 0$$

Electrostatic Potential Due to Ring on an Axial Point

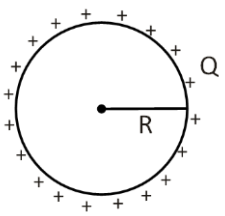
Part - 06

Electrostatic potential due to a uniformly charged ring:

(i) At the centre -

Consider a small element on the ring having charge dq then potential due to this at centre will be

$dV = \frac{kdq}{R}$



so potential due to complete ring

$V = \int dv$

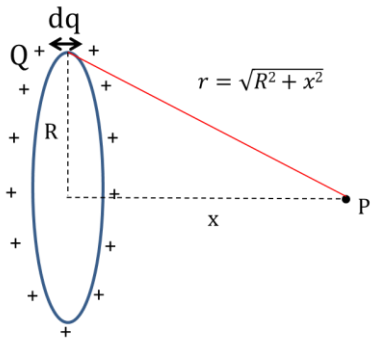
$\Rightarrow V = \int_0^Q \frac{kdq}{R}$

$\Rightarrow V = \frac{kQ}{R}$

(ii) On its axis-

Let electric potential at point P due to a small element of length $d\ell$ be dV, then

$dV = \frac{dq}{4\pi\varepsilon_0 r}$ (distance between small element and point P is equal to r)



$\therefore$ Total potential due to the ring on its axis : $V = \int dV = \int \frac{dq}{4\pi\varepsilon_0 r} = \frac{1}{4\pi\varepsilon_0 r}\int dq$

$V = \frac{Q}{4\pi\varepsilon_0 (R^2 + x^2)^{1/2}} \Rightarrow \boxed{V = \frac{kQ}{\sqrt{R^2 + x^2}}}$

Putting x = 0 and potential at the centre of ring, $V_{center} = \frac{kQ}{R}$

Illustration 1 :

Find potential on the axis of a ring (Q,R) where its field is maximum?

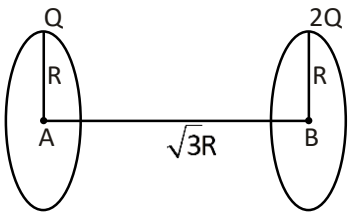
Solution:

As we know that field due to a charged ring is maximum at x = $\pm\frac{R}{\sqrt{2}}$ from the centre on its axis, so potential at these points will be –

$$V=\frac{kQ}{\sqrt{R^2+\left(\frac{R}{\sqrt{2}}\right)}}=\frac{kQ}{\sqrt{R^2+\frac{R^2}{2}}}=\frac{kQ}{\sqrt{\frac{3R^2}{2}}}=\frac{\sqrt{2}kQ}{\sqrt{3}R}$$

Illustration 2 :

Find $V_A - V_B$ for the system of charged rings placed coaxially as shown -



Solution:

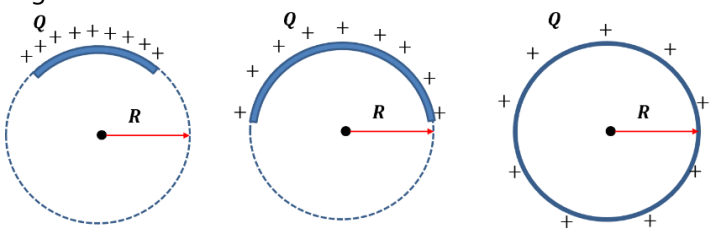
Using formula we can write -

$$V_A=\frac{kQ}{R}+\frac{2kQ}{2R}=\frac{2kQ}{R}$$

$$V_B=\frac{2kQ}{R}+\frac{kQ}{2R}=\frac{5kQ}{2R}$$

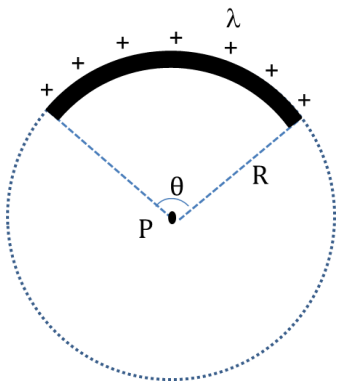
$$\therefore\ V_A-V_B=\frac{2kQ}{R}-\frac{5kQ}{2R}=-\frac{kQ}{2R}$$

Note : Consider following charge distributions :



Here, distance of total charge from Centre of ring is R in all cases so potential at Centre in every case will be $V=\frac{kQ}{R}$

Electrostatic potential due to a uniformly charged circular arc –

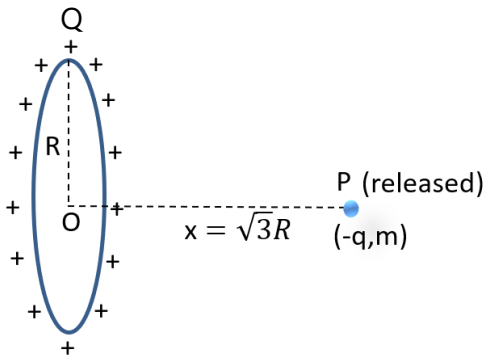


Here, linear charge density of arc is λ so total charge of the area can be written as $Q = \lambda R\theta$ hence potential will be

$$V = \frac{kQ}{R} = \frac{k(\theta R\lambda)}{R} = k\theta\lambda$$

Illustration 3 :

Find speed of the charge particle, when it reaches at centre of the ring from point P on the axis as shown.



Solution:

Using COME between points P and O -

$(KE)_i + (PE)_i = (KE)_f + (PE)_f$

$$\Rightarrow 0 - q\frac{kQ}{2R} = \frac{1}{2}mv^2 - q\frac{kQ}{R}$$

$$\Rightarrow \frac{kQq}{2R} = \frac{1}{2}mv^2$$

$$\Rightarrow v = \sqrt{\frac{kQq}{mR}}$$

Illustration 4 :

A ball of mass 4g has 2μC of charge. It is released from the position where potential is 60 Volt then find its speed when reaches the position where potential is 20 Volt.

Solution:

Using COME between initial point and final position

$(KE)_i + (PE)_i = (KE)_f + (PE)_f$

$$\Rightarrow 0 + qV_i = \frac{1}{2}mv^2 + qV_f$$

$$\Rightarrow 0 + (2\mu C)60 = \frac{1}{2}mv^2 + (2\mu C)20$$

$$\Rightarrow 80\times10^{-6} = \frac{1}{2}\left(4\times10^{-3}\right)v^2$$

$$\Rightarrow v^2 = 4\times10^{-2}$$

$$\Rightarrow v = 0.2 \text{ m/s}$$

Illustration 5 :



Work done by external agent in moving an electron and a proton from A to B and from B to A.

Solution:

For electron	For proton
$W_{(A\to B)} = -e(300V - 600V)$	$W_{(A\to B)} = +e(300V\text{-}600V)$
$W_{(A\to B)} = 300eV$	$W_{(A\to B)} = -300eV$
$W_{(B\to A)} = -e(600V - 300V)$	$W_{(B\to A)} = +e(600V - 300V)$
$W_{(B\to A)} = -300eV$	$W_{(B\to A)} = +300eV$

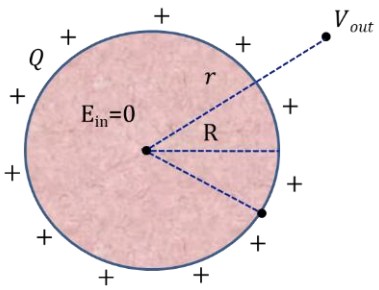
Note : *Think about work done by field also.*

Electrostatic Potential Due to Sphere and Spherical Shell

Part - 07

Electrostatic Potential due to a Charged Conducting Sphere or Spherical Shell:

Surface charge density $\sigma = \dfrac{Q}{4\pi R^2}$



(i) For outside points (r > R)

∵ Whole charge is supposed to be at centre so

$$\int_0^V dV = -\int_\infty^r E.dr$$

$$V = \int_\infty^r \frac{KQ}{r^2} dr$$

$$V = \frac{KQ}{r} = \frac{Q}{4\pi\varepsilon_0 r}$$

$$V = \frac{Q}{4\pi R^2}\varepsilon_0 r \Rightarrow V = \frac{KQ}{r} = \frac{\sigma R^2}{\varepsilon_0 r}$$

($Q \to \sigma$, R^2)

(ii) For points on the surface (r = R)

$$V = \frac{KQ}{R} = \frac{Q}{4\pi\varepsilon_0 R} \Rightarrow V = \frac{Q}{4\pi R^2} \times \frac{R}{\varepsilon_0} \Rightarrow V = \frac{\sigma R}{\varepsilon_0} = \frac{KQ}{R}$$

(iii) For Inside points (r < R)

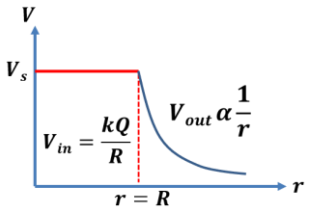
∵ For all points inside the sphere E_{in} is same so

$$V = \int_0^V dV = -\int E.dr$$

$$\int_0^V dV = -\int_\infty^R \vec{E}.\vec{dr} - \int_R^r \vec{E}.\vec{dr} \quad \Rightarrow \quad V = -\int_\infty^R \frac{KQdr}{r^2} - \int_R^r Edr \quad \Rightarrow \quad V = \frac{KQ}{R}$$

Variation of potential with r

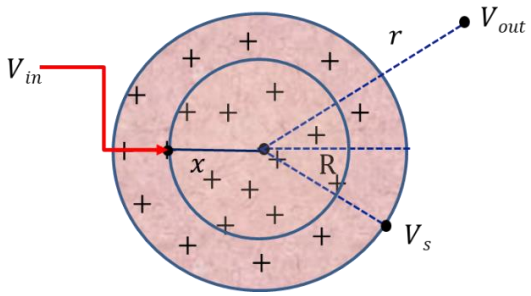
$V_{out} = \frac{kQ}{r}$ and $V_S = V_{in} = \frac{kQ}{R}$



Electric Potential due to Uniformly Charged Non-conducting Sphere

(i) Outside (r > R)

Because sphere will act like charge Q kept at centre.



$$\int_0^V dV = -\int_\infty^r \vec{E}.dr$$

$$V = -\int_\infty^r \frac{KQ}{r^2} dr \qquad V = \frac{KQ}{r}$$

(ii) On the surface (r = R)

Putting r = R in above equation we get -

V surface = $\frac{KQ}{R}$

(iii) Inside (r < R) (Here, x = r for this case in above diagram)

$$\int dV = -\int_\infty^R \vec{E}.\vec{dr} - \int_R^r \vec{E}.\vec{dr}$$

$$\Rightarrow V = -\int_\infty^R \frac{KQ}{r^2} dr - \int_R^r \frac{KQr}{R^3} dr \quad \Rightarrow V = \frac{KQ}{R} - \frac{KQ}{R^3}\left(\frac{r^2}{2} - \frac{R^2}{2}\right) \quad \Rightarrow \frac{KQ}{R} - \frac{KQ}{2R^3}(r^2 - R^2)$$

$$\Rightarrow V = \frac{KQ}{2R^3}(2R^2 - (r^2 - R^2)]$$

$$\Rightarrow \text{Vin} = \frac{KQ}{2R^3}\left[3R^2 - r^2\right]$$

At center, r = 0

$$\Rightarrow V_{center} = \frac{3}{2}\frac{KQ}{R} = \frac{3}{2}V_{surface}$$

Variation of potential with r

$$V_{in} = \frac{kQ}{2R^3}\left(3R^2 - x^2\right); \quad V_s = \frac{kQ}{R}; \quad V_{out} = \frac{kQ}{r}$$

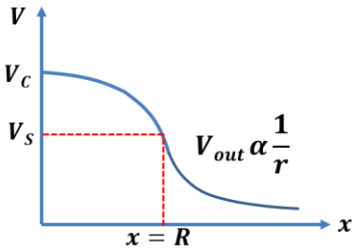


Illustration 1 :

A conducting sphere of radius R has charge Q. Find potential at distance of (a) $\frac{R}{2}$ (b) $\frac{3R}{2}$ from centre.

Solution:

(a) r < R i.e. inside the surface $V_{in} = V_S = \frac{kQ}{R}$

(b) r > R i.e. outside the surface $V_{out} = \frac{kQ}{r} = \frac{2kQ}{3R}$

Illustration 2 :

A nonconducting uniformly charged sphere of radius R has charge Q. Find potential at distance of (a) $\frac{R}{2}$ (b) $\frac{3R}{2}$ from centre.

Solution:

(a) r < R i.e. inside the surface $V_{in} = \frac{kQ}{2R^3}\left(3R^2 - r^2\right)$ Put $r = \frac{R}{2} \Rightarrow \text{Vin} = \frac{11kQ}{8R}$

(b) r > R i.e. outside the surface $V_{out} = \frac{kQ}{r}$

$$V_{out} = \frac{kQ}{r} = \frac{2kQ}{3R}$$

Illustration 3 :

A conducting sphere of radius R has electric field E at distance 3R from centre. Find potential difference between centre of the sphere & the point where electric field is E.

Solution:

$$E = \frac{kQ}{(3R)^2} = \frac{kQ}{9R^2} \Rightarrow kQ = 9ER^2$$

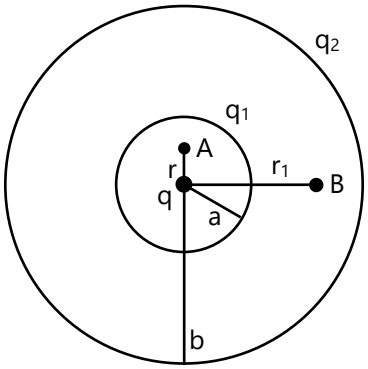
$$V_{centre} = \frac{kQ}{R} = \frac{9ER^2}{R} = 9ER$$

$$V_P = \frac{kQ}{r} = \frac{9ER^2}{3R} = 3ER$$

$$V_{centre} - V_P = 9ER - 3ER = 6ER$$

Problems on potential due to concentric spherical shells -

Illustration 4 :



Find potential at points A and B due to given arrangement of shells.

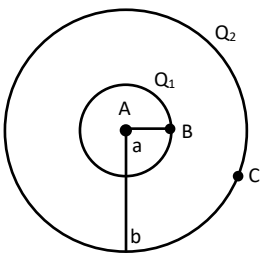
Solution:

$$V_A = \frac{kq}{r} + \frac{kq_1}{a} + \frac{kq_2}{b}$$

$$V_B = \frac{kq}{r_1} + \frac{kq_1}{r_1} + \frac{kq_2}{b}$$

Illustration 5 :

Find potential at points A,B and C due to given arrangement of shells.



Solution:

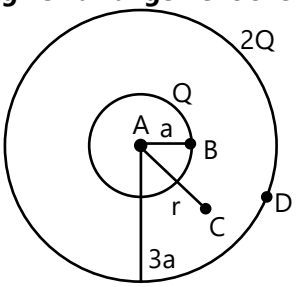
$$V_A = \frac{kQ_1}{a} + \frac{kQ_2}{b}$$

$$V_B = \frac{kQ_1}{a} + \frac{kQ_2}{b}$$

$$V_C = \frac{kQ_1}{b} + \frac{kQ_2}{b}$$

Illustration 6 :

Find potential at points A,B,C and D due to given arrangement of shells.



Solution:

$$V_A = \frac{kQ}{a} + \frac{2kQ}{3a}$$

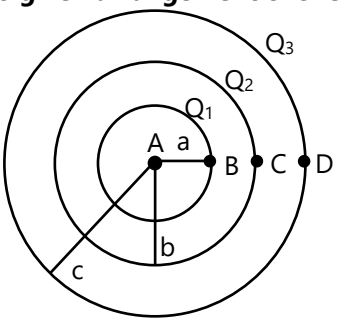
$$V_B = \frac{kQ}{a} + \frac{2kQ}{3a}$$

$$V_C = \frac{kQ}{r} + \frac{2kQ}{3a}$$

$$V_D = \frac{kQ}{3a} + \frac{2kQ}{3a}$$

Illustration 7 :

Find potential at points A,B,C and D due to given arrangement of shells.



Solution:

$$V_A = \frac{kQ_1}{a} + \frac{kQ_2}{b} + \frac{kQ_3}{c}$$

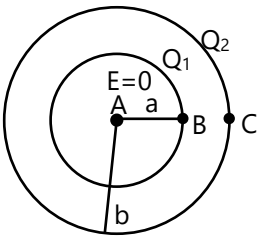
$$V_B = \frac{kQ_1}{a} + \frac{kQ_2}{b} + \frac{kQ_3}{c}$$

$$V_C = \frac{kQ_1}{b} + \frac{kQ_2}{b} + \frac{kQ_3}{c}$$

$$V_D = \frac{kQ_1}{c} + \frac{kQ_2}{c} + \frac{kQ_3}{c}$$

Illustration 8 :

Find V_A, V_B and V_C in the given arrangement of shells



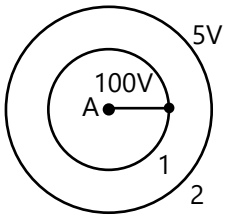
Solution:

$$V_A = V_B = \frac{kQ_1}{a} + \frac{kQ_2}{b}$$

$$V_C = \frac{kQ_1}{b} + \frac{kQ_2}{b}$$

Illustration 9:

The value of potential at A is

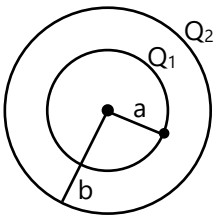


Solution:

Potential of A will be 100V because A is within sphere 1 which is having potential 100V at its surface so within it potential at everywhere will be 100V only.

Illustration 10:

Find V_{in} - V_{out} for the given arrangement of shells.



Solution:

$$V_{in} = \frac{kQ_1}{a} + \frac{kQ_2}{b}$$

$$V_{out} = \frac{kQ_1}{b} + \frac{kQ_2}{b}$$

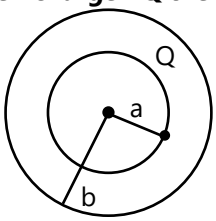
$$V_{in} - V_{out} = \frac{kq_1}{a} - \frac{kQ_2}{b}$$

$$V_{in} - V_{out} = kQ_1\left[\frac{1}{a} - \frac{1}{b}\right]$$

Note : *P.D. between 2 concentric spherical shalls depends only on the charge of inner sphere.*

Illustration 11 :

Initially V_{in} - V_{out} = V now if outer sphere is given charge 4Q then V_{in} - V_{out} will be?

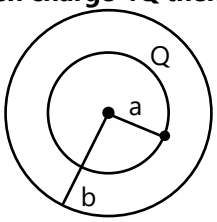


Solution:

Potential difference between them will not depend upon charge on outer surface so $V_{in} - V_{out}$ will be V only.

Illustration 12 :

Initially $V_{in} - V_{out} = V$ now if inner sphere is given charge 4Q then $V_{in} - V_{out}$ will be?



Solution:

Potential difference between them is directly proportional to charge on inner sphere so $V_{in} - V_{out}$ will be 4V.

Potential Difference, Equipotential Surfaces, Relation between Electric Field and Potential

Part -08

Equipotential Surfaces (E.P.S.)

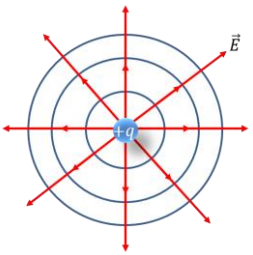
Locus of all such points which are at same potential or the surface at which potential of all the points are equal is known as equipotential surface (EPS). It can be of any shape.

Properties of equipotential surface

- Electric field intensity is always perpendicular to EPS.
- Work done in moving a charge from one point of EPS to other point of same EPS is always zero.
- Two EPS can never intersect each other.

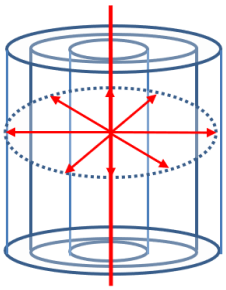
(1) EPS due to point charge -

EPS of point charge is spherical in shape.



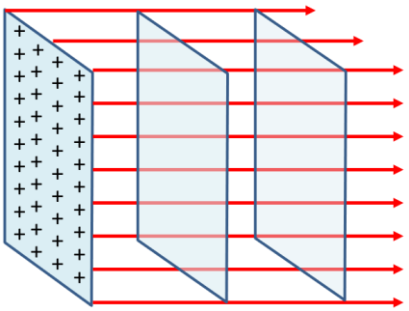
(2) EPS due to line charge –

EPS of line charge is cylindrical in shape.

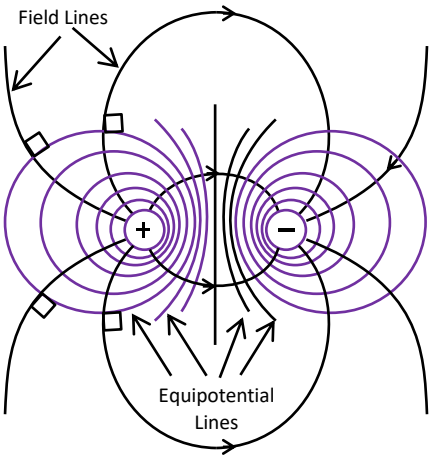


(3) EPS due to large plane charged sheet –

EPS of large plane sheet is plane in shape.



(4) EPS due to unlike charges -



(5) EPS due to like charges -

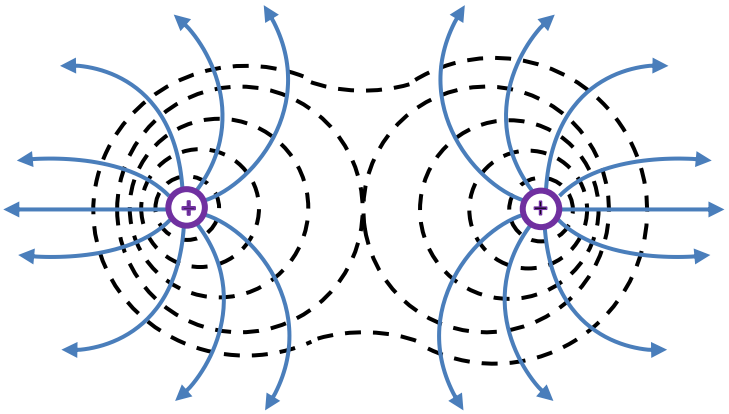
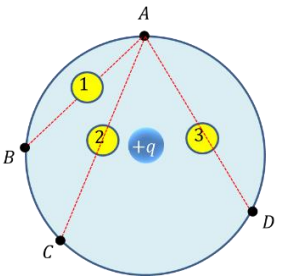


Illustration 1 :

A charge in moving via three paths, 1, 2, and 3. as shown in figure. Find work done in each case.



Solution:

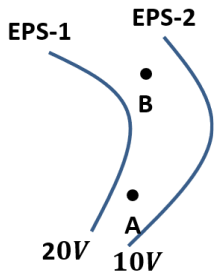
Potential at points A, B, C and D is

$$V = \frac{kq}{R}$$

All the points are equipotential points, so work done in each case is zero.

Illustration 2 :

Find relation between E_A and E_B.

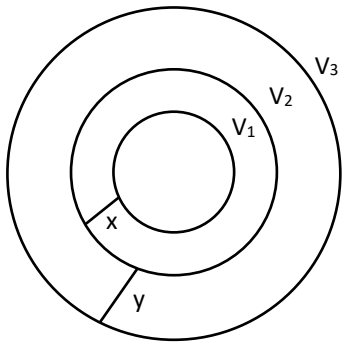


Solution:

$\because \quad |E| = \frac{\Delta V}{d}$ and here ΔV is same, but d is less near point A so E_A will be greater than E_B.

Illustration 3:

Concentric spherical equipotential surfaces due to a point charge are shown in the diagram, if $V_1 - V_2 = V_2 - V_3$ then -



Solution:

If electric field between V_1 and V_2 be the E_1 and between V_2 and V_3 be the E_2 then we can write (using approximation)

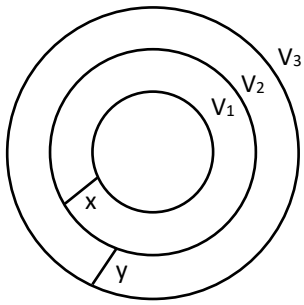
$$V_1 - V_2 = E_1(x)$$

$$V_2 - V_3 = E_2(y)$$

We can see here $E_1 > E_2$ so according to given condition that $V_1 - V_2 = V_2 - V_3$ we can conclude that $x<y$.

Illustration 4:

Concentric spherical equipotential surfaces due to a point charge are shown in the diagram, if x = y then find relation between $(V_1 - V_2)$ and $(V_2 - V_3)$.



Solution:

We can write (using approximation)

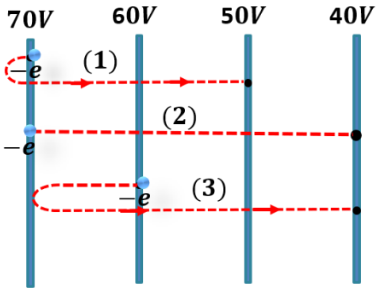
$V_1 - V_2 = E_1(x)$

$V_2 - V_3 = E_2(y)$

We know that, Here $E_1 > E_2$ and given x = y so we can conclude that $(V_1 - V_2) > (V_2 - V_3)$.

Illustration 5 :

An electron is moved via 3 different paths through various equipotential lines. Arrange them in deceasing order of work done.



Solution:

Using $W = q\,\Delta V = q(V_f - V_i)$, we have

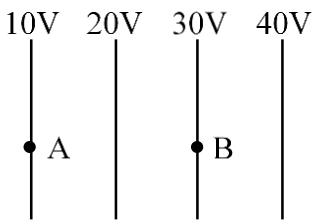
$W_{(path\text{-}1)} = -e(50 - 70) = +20eV$

$W_{(path\text{-}2)} = -e(40 - 70) = +30eV$

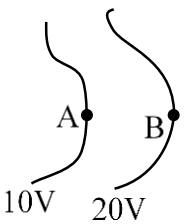
$W_{(path\text{-}3)} = -e(40 - 60) = +20eV$

Illustration 6 :

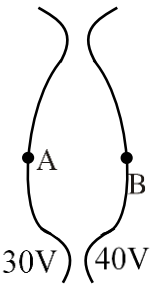
Find which has more work done in moving a charge from A to B?



(1)



(2)



(3)

Solution:

Use $W = q(\Delta V)$

$W_1 > W_2 = W_3$

Relation between electrostatic field (E) and potential (V) :

We know that $$F = -\frac{dU}{dr}$$

Dividing both sides by q

$$\Rightarrow \frac{F}{q} = -\frac{d}{dr}\left(\frac{U}{q}\right)$$

$\Rightarrow \; E = -\dfrac{dV}{dr}$

As we know $\vec{F} = q\vec{E}$ & U = qV

$q\vec{E} = -\dfrac{d}{dr}(qV)$ i.e $q\vec{E} = -$ gradient (qV)

Or $\vec{E} = -$ gradient (V)

I.e. negative gradient of electric potential is equal to electric field.

Since electric field is negative gradient of potential, thus potential decreases in the direction of $\vec{E}$.

Finding E when V is given :

1. Gradient → Rate of change with distance $\Rightarrow E = \dfrac{-dV}{dr}$ or $\vec{E} = \dfrac{-dV}{dr}\hat{r}$

2. In cartesian system

   As $\vec{E} = -\nabla V$ and $\nabla = \dfrac{\partial}{\partial x}\hat{i} + \dfrac{\partial}{\partial y}\hat{j} + \dfrac{\partial}{\partial z}\hat{k}$

   $\vec{E} = -\dfrac{\partial V}{\partial x}\hat{i} - \dfrac{\partial V}{\partial y}\hat{j} - \dfrac{\partial V}{\partial z}\hat{k}$

   Electric field in the space is -

   $\vec{E} = E_x\hat{i} + E_y\hat{j} + E_z\hat{k}$

   Here, $E_x = -\dfrac{\partial V}{\partial x}$; $E_y = -\dfrac{\partial V}{\partial y}$; $E_z = -\dfrac{\partial V}{\partial z}$ and

   $\dfrac{\partial}{dx} \rightarrow$ partial differentiation with respect to x, keeping y & z constant.

   $\dfrac{\partial}{dy} \rightarrow$ partial differentiation with respect to y, keeping x & z constant.

   $\dfrac{\partial}{dz} \rightarrow$ partial differentiation with respect to z, keeping x & y constant.

Finding V when E is given:

As we know

$\because E = -\dfrac{dV}{dr} \;\Rightarrow\; dV = -E\,dr \;\Rightarrow dV = -\vec{E}\,.\,\vec{dr}$

On integration

$\Delta V = V_P - V_{ref} = -\int_{ref}^{P} \vec{E}.\vec{dr}$ or $V_P - V_{ref} = -\int_{X_{ref}}^{X_P} E_x dx - \int_{Y_{ref}}^{Y_P} E_y dy - \int_{Z_{ref}}^{Z_P} E_Z dz$.

Illustration 7 :

If $V=8x^2$, find electric field in the region.

Solution:

$E_x = -\dfrac{\partial V}{\partial x} = -16x$

$\vec{E} = E_x\hat{i} + E_y\hat{j} + E_z\hat{k}$ (E_y and E_z is zero)

$\Rightarrow \; \vec{E} = -16x\hat{i}$

$\Rightarrow \; \vec{E} = 16x(-\hat{i})$

Illustration 8 :

If $V = x^2y - yz^3 - 3x^2yz$, find x, y and z components of electric field.

Solution:

$$E_x = -\frac{\partial V}{\partial x} = -\left[y(2x) - 0 - 3yz(2x)\right]$$

$E_x = (-2xy + 6xyz)$

$$E_y = -\frac{\partial V}{\partial y} = -\left[x^2(1) - z^3(1) - 3x^2(1)z\right]$$

$E_y = (-x^2 + z^3 + 3x^2 z)$

$$E_z = -\frac{\partial V}{\partial z} = -\left[0 - y(3z^2) - 3x^2y(1)\right]$$

$E_z = (3yz^2 + 3x^2y)$

Illustration 9 :

If $V=x^2y + y^2xz + 5$ then find $\vec{E}$ at (1,1,0).

Solution:

$$E_x = -\frac{\partial V}{\partial x} = -\left[2xy + y^2z\right]$$

$$E_y = -\frac{\partial V}{\partial y} = -\left[x^2 + 2yxz\right]$$

$$E_z = -\frac{\partial V}{\partial Z} = -\left[0 + y^2x\right]$$

At (1,1,0) $\Rightarrow E_x = -2\hat{i}$; $E_y = -\hat{j}$; $E_z = -\hat{k} \Rightarrow \vec{E} = -2\hat{i} - \hat{j} - \hat{k}$

Illustration 10 :

If $V = x^2y - yz^3 - 3x^2yz + 3x - 4y - 7$, find electric field at origin.

Solution:

$$E_x = -\frac{\partial V}{\partial x} = -\left[2xy - 0 - 6xyz + 3\right]$$

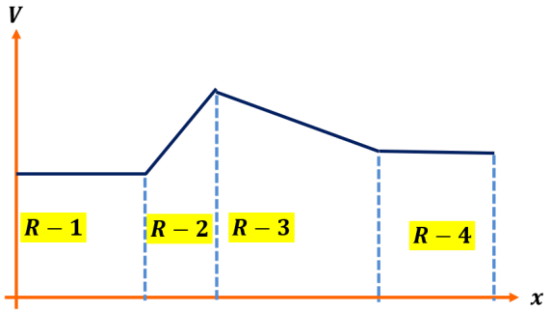
$$E_y = -\frac{\partial V}{\partial y} = -\left[x^2 - z^3 - 3x^2z - 4\right]$$

$$E_z = -\frac{\partial V}{\partial Z} = -\left[0 - 3yz^2 - 3x^2y\right]$$

At (0,0,0) $\Rightarrow E_x = -3\hat{i}$; $E_y = 4\hat{j}$; $E_z = 0 \Rightarrow \vec{E} = -3\hat{i} + 4\hat{j}$

Illustration 11 :

In which region electric field is maximum?



Solution:

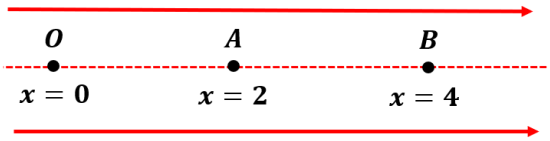
$$E = -\frac{dV}{dr} = \text{slope}$$

So, Region R – 2 have maximum electric field.

Illustration 12 :

If $\vec{E} = \frac{40}{x^2}(\hat{i})\ N/m$, then find potential difference between the points x = 2m and x = 4m.

Solution:



$$\int dV = -\int E dx \cos\theta$$

$$\Rightarrow \int_{V_A}^{V_B} dV = -\int_{x=2}^{x=4} E dx \cos 0°$$

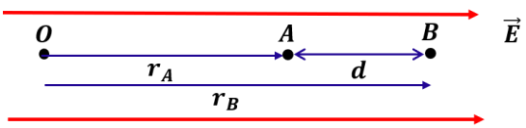
$$\Rightarrow (V_B - V_A) = -\int_{x=2}^{x=4} \frac{40}{x^2} dx$$

$$\Rightarrow (V_B - V_A) = -40\left[-\frac{1}{x}\right]_2^4$$

$$\Rightarrow (V_B - V_A) = +40\left[\frac{1}{4} - \frac{1}{2}\right] = -\frac{40}{4} = -10V$$

$\Rightarrow (V_A - V_B) = +10V$

Note : In uniform electric field -



$$\int dV = -\int E dr \cos\theta$$

$$\Rightarrow \int_{V_A}^{V_B} dV = -E\int_{r_A}^{r_B} dr$$

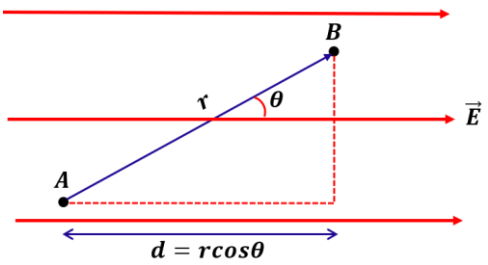
$\Rightarrow (V_B - V_A) = -E(r_B - r_A)$

$\Rightarrow$ $(V_A - V_B) = + E(r_B - r_A)$

$\Rightarrow$ $\Delta V = Ed$

P.D. = E(distance between two points along electric field)

Illustration 13 :



Solution:

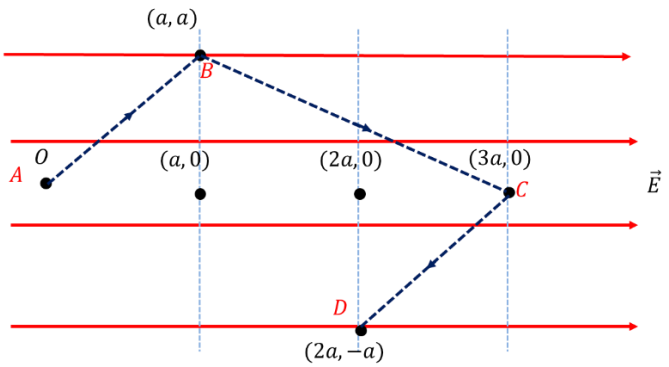
P.D.= E(distance between two points along electric field)

$\Rightarrow$ $\Delta V = Ed = E r\cos\theta$

Illustration 14 :

There is a uniform electric field parallel to x axis as shown. Find work done by the electric field in moving a charge +q from

(a) A to B (b) B to C (c) A to D



Solution:

$W_{(A\to B)} = +qE[0 - (-a)] = +qEa$

$W_{(B\to C)} = +qE[(-a) - (-3a)] = +2qEa$

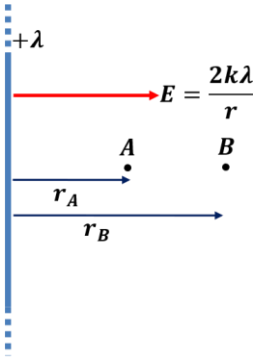
$W_{(A\to D)} = +qE[0 - (-2a)] = +2qEa$

Note : *In conservative field work done depends only on initial and final positions, does not depend on path followed.*

Electrostatic Potential difference for line charge and sheet -

Net potential at a point due to infinite geometry cannot be obtained but potential difference between two nearby points can be obtained.

(i) For infinite line charge -



For nonuniform electric field :

$$\int_{V_B}^{V_A} dV = -\int_{r_B}^{r_A} \vec{E}.\vec{dr}$$

$$\Rightarrow V_A - V_B = -\int_{r_B}^{r_A} \frac{2k\lambda dr}{r}$$

$$\Rightarrow (V_A - V_B) = 2k\lambda \ln\left(\frac{r_B}{r_A}\right)$$

(ii) For large plane sheet -

Here electric field is uniform so,

$(V_A - V_B) = Ed$

$$(V_A - V_B) = \frac{\sigma}{2\epsilon_0}(r_B - r_A)$$

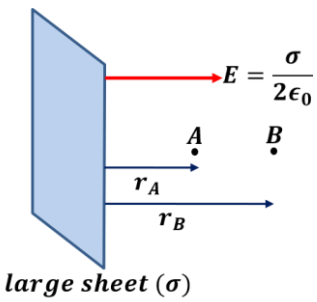
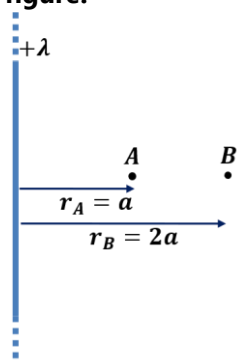


Illustration 15 :

Find p.d. between points A and B as shown in figure.



Solution:

From $(V_A - V_B) = 2k\lambda \ln\left(\frac{r_B}{r_A}\right)$

$(V_A - V_B) = 2k\lambda \ln\left(\frac{2a}{a}\right)$

$(V_A - V_B) = 2k\lambda \ln(2)$

Crux Pan

- If potential energy at infinity is zero, external work done in changing the configuration of a system from 'i' to 'f' is $W = U_f - U_i$.
- In assembling a given charge system ($U_i = 0$) $\Rightarrow W = U_f$ and in disassembling ($U_f = 0$) $\Rightarrow W = -U_i$
- eV is the smallest practical unit of energy used in atomic and nuclear physics. (1 eV = 1.6×10^{-19} J)
- Potential is theoretically zero at infinity; practically we consider it zero on the earth surface.
- Potential on earth is assumed to be zero as it is a large conductor and its potential is more or less constant in relation to finite charge given to or taken from it.
- The direction of electric field is from regions of higher electric potential to regions of lower potential irrespective of the charge configuration.
- If one moves in the direction of $\vec{E}$, electric potential V decreases; If one moves in the direction opposite to $\vec{E}$ potential increases.
- Electric potential is the same at all points in a region where E = 0.
- Equipotential surfaces need not be physical surfaces. Any imaginary surface over which the electric field is perpendicular to it everywhere is called an equipotential surface.
- Equipotential surfaces are closely spaced where electric field intensity is large and widely spaced where electric field intensity is small.
- Electric field must be zero within an equipotential volume.
- Different equipotential surfaces due to a point charge whose potentials differ by a constant amount (say 1 volt) are not equispaced.
- Two equipotential surfaces never intersect.
- When we place a charged conductor inside a hollow conductor and connect them by a conducting wire, the charge will be completely transferred from the inner conductor to the other conductor.

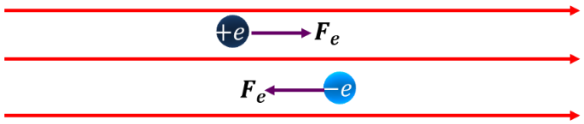
Motion of Charged Particle in Uniform Electric Field

Part - 09

Motion of a charge particle in uniform electric field

Case – I : When a charged particle is released from rest (i.e. u = 0)

Consider an electron and a proton both are released in uniform electric field, they travelled along straight line in opposite direction with constant acceleration.



$a = \frac{eE}{m}$ = constant, hence equations of motion can be applied.

Also, $F_{proton} = F_{electron}$ but, $\vec{F}_{proton} = -\vec{F}_{electron}$ and

$\vec{a}_{proton} \neq -\vec{a}_{electron}$

$a_e = \frac{eE}{m_e}$ and $a_p = \frac{eE}{m_p}$

$\frac{a_e}{a_p} = \frac{m_p}{m_e} = 1837$ {As $m_p = 1837 m_e$}

Distance travelled by charge particle in uniform electric field.

Using $s = ut + \frac{1}{2}at^2 \Rightarrow s = \frac{1}{2}at^2$ or $t = \sqrt{\frac{2s}{a}} = \sqrt{\frac{2sm}{eE}}$

Here t is time taken by charge particle to move distance s in uniform electric field.

Illustration 1 :

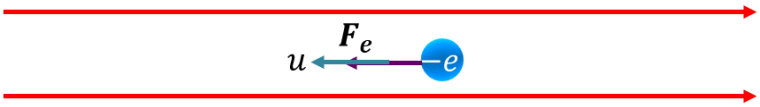
An electron is allowed to fall through some distance in uniform electric field in time t_1. Now the direction of electric field is reversed and proton is allowed to fall through same distance in time t_2. Then relation between t_1 and t_2.

Solution:

In case (1), $s = \frac{1}{2}a_e t_1^2$ and In case (2), $s = \frac{1}{2}a_p t_2^2$

$\frac{t_1}{t_2} = \sqrt{\frac{a_p}{a_e}} = \sqrt{\frac{m_e}{m_p}} = \frac{1}{\sqrt{1837}} \Rightarrow t_1 < t_2$

(Case-II) If the electron is projected opposite to the direction of electric field



In this case electron will accelerate as force is in the direction of motion so distance travelled by electron in time t will be

$$s = ut + \frac{1}{2}\left(\frac{eE}{m}\right)t^2$$

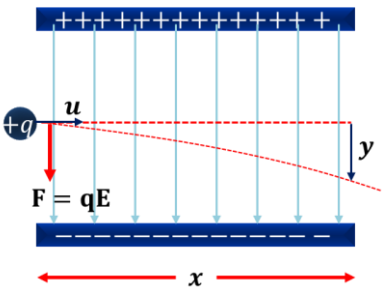
Velocity of electron after time t will be

$$v = u + at \quad \Rightarrow V = u + \left(\frac{eE}{m}\right)t$$

Case –III : When the charge particle is projected perpendicular to a uniform electric field $\left(\vec{u} \perp \vec{E}\right)$ as shown

$u_x = u$; but $a_x = 0$

$u_y = 0$; but $a_y = \frac{qE}{m}$ = constant



along x-direction

$u_x = u$

$a_x = 0$

$v_x = u_x + a_x t$

$v_x = u$

$x = u_x t + \frac{1}{2} a_x t^2$

$x = ut$

along y-direction

$u_y = 0$

$a_y = \frac{qE}{m}$ =constant

$v_y = u_y = a_y t$

$v_y = \frac{qEt}{m}$

$y = a_y t + \frac{1}{2} a_y t^2$

$y = \frac{1}{2}\frac{qE}{m} t^2$

Deviation of the particle from its path

$$y = \frac{1}{2}\frac{qE}{m} t^2$$

Trajectory equation (x & y relation)

$$y = \frac{1}{2}\frac{qE}{m} t^2 = \frac{1}{2}\frac{qE}{m}\left(\frac{x}{u}\right)^2$$

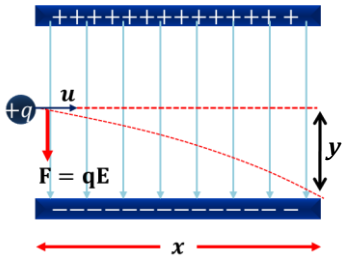
[from $x = ut \Rightarrow t = \frac{x}{u}$]

$$y = \left(\frac{1}{2}\frac{qE}{mu^2}\right)x^2 \quad \Rightarrow \quad y = kx^2 \text{ (parabolic path)}$$

Also, deviation –

$$y = \frac{1}{2}\left(\frac{qE}{mu^2}\right)x^2 = \left(\frac{qE}{4K_i}\right)x^2$$

(Where $K_i = \frac{1}{2}mu^2$ & x = length of the plates)



Velocity of particle at time 't'

$$\vec{v} = v_x\hat{i} + v_y\hat{j}$$

$$\vec{v} = u\hat{i} + \left(\frac{qE}{m}t\right)\hat{j}$$

Speed, $v = \sqrt{v_x^2 + v_y^2}$ and in diagram $\tan\alpha = \frac{v_y}{v_x}$

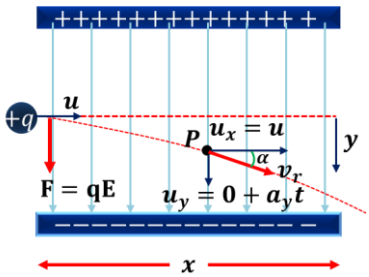
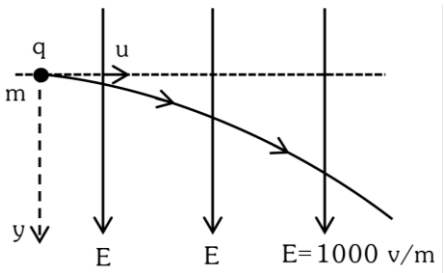


Illustration 2 :

A particle of mass 1g and charge 10μC is projected with velocity 30 m/s in direction perpendicular to a uniform electric field of 1000 V/m. Find (a) its speed after 4 second (b) deviation after 4 second.

Solution:



(a) v_x = u = 30 m/s ; $v_y = \left(\frac{qE}{m}\right)t$

$$v_y = \left(\frac{10\times10^{-6}\times10^3}{10^{-3}}\right)\times4 \Rightarrow v_y = 40\,m/s$$

Speed after 4 second

$$c = \sqrt{v_x^2 + v_y^2} = \sqrt{30^2 + 40^2} = 50\,m/s$$

(b) Deviation → y

$$y = \frac{1}{2}\left(\frac{qE}{m}\right)t^2 = \frac{1}{2}\times10\times4^2 \Rightarrow y = 80m$$

Illustration 3 :

An electron enters into a uniform electric field of 364 V/m with a velocity 100 m/s in a direction perpendicular to the electric field. If it travels a distance of 0.01 mm in the direction of electric field, find deviation in its path:

Solution:

Here $y = \frac{qEx^2}{2mu^2}$

q = e = 1.6×10^{-19} C ; E = 364 V/m

m = 9.1×10^{-31} kg ; u = 100 m/s

and x = 0.01 mm = 10^{-5} m

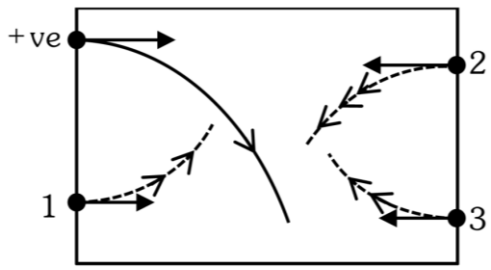
$$y = \frac{1.6\times10^{-19}\times364\times(10^{-5})^2}{2\times9.1\times10^{-31}\times(10^2)^2}$$

$$y = \frac{1.6\times364}{18.2}\times10^{-2}$$

Y = 32×10^{-2} m or y = 0.32 m

Illustration 4 :

Find nature of charge of particles 1, 2 and 3



Solution:

Since $\vec{E}$ is in downward direction

So 1 → -ve; 2→ +ve ; 3→ -ve

Illustration 5 :

A charge particle Q (q,m) is released from rest in uniform field $\vec{E}$. Then find –

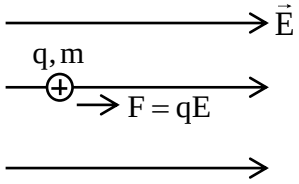
(a) velocity after t sec.

(b) displacement after t sec.

(c) KE after t sec.

(d) rate of charge in KE at t sec.

Solution:



(a) For particle $u = 0,\ a = \frac{qE}{m}$

So Velocity after t sec, $v = u + at$

$$\Rightarrow v = 0 + \left(\frac{qE}{m}\right)t = \frac{qE}{m}t$$

(b) displacement after t sec , $s = ut + \frac{1}{2}at^2$

$$\Rightarrow s = 0 + \left(\frac{qE}{2m}\right)t^2 = \left(\frac{qE}{2m}\right)t^2$$

(c) KE after t sec =

$$= \frac{m}{2}\left(\frac{qE}{m}t\right)^2$$

$$= \frac{q^2E^2t^2}{2m}$$

(d) Rate of change of KE at t sec = $\frac{d}{dt}(KE)_t$

$$= \frac{d}{dt}\left(\frac{q^2E^2t^2}{2m}\right)$$

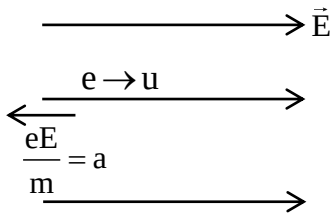
$$= \frac{q^2E^2}{2m}(2t)$$

$$= \frac{q^2E^2t}{m}$$

Illustration 6 :

An electron is projected with speed u along a uniform electric field E, then find distance travelled before coming to rest.

Solution:



For electron $a = \frac{-eE}{m}$ and $v_{final} = 0$

If it comes to rest after t time,

Putting values in $v = u + at$

$$0 = u - \frac{Ee}{m} t$$

$$\Rightarrow \quad t = \frac{um}{Ee}$$

Distance covered $S = ut + \frac{1}{2}at^2$

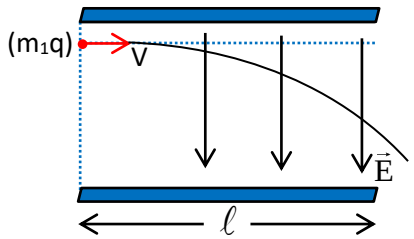
$$\Rightarrow S = u\left(\frac{mu}{eE}\right) - \frac{1}{2}\left(\frac{eE}{m}\right)\left(\frac{mu}{eE}\right)^2$$

$$\Rightarrow S = \frac{mu}{eE}\left[u - \frac{1}{2}\left(\frac{eE}{m}\right)\left(\frac{mu}{Ee}\right)\right]$$

$$\Rightarrow S = \frac{mu^2}{2eE} = \frac{K_i}{eE}$$

Illustration 7 :

Charge (q,m) is projected with speed V in uniform transverse $\vec{E}$ confined in a region of length ℓ as shown in figure.



(1) Time for which particle remains in $\vec{E}$

(2) v of particle when it leaves $\vec{E}$

(3) Deviation of particle

Solution:

(1) Till the time particle covers horizontal distance, ℓ it will remain in $\vec{E}$.

For particle $u_x = v$ (initial speed), $a_x = 0$, $S_x = \ell$

$$S = ut + \frac{1}{2}at^2$$

$$\ell = vt + 0 \Rightarrow \quad t = \frac{\ell}{V}$$

(2) Speed of particle where it leaves $v_{final} = u + at$

$v_{x(final)} = v$ & $v_{y(final)} = u_y + a_y t$

$$v_y = 0 + \left(\frac{qE}{m}\right)\frac{\ell}{v}$$ so final speed = $\sqrt{v_x^2 + v_y^2}$

Final speed $v_{(final)} = \sqrt{v^2 + \left(\frac{qE\ell}{mv}\right)^2}$

(3) Deviation (vertically) in time $t = \frac{\ell}{v}$

$$S_y = u_y t + \frac{1}{2}at^2$$

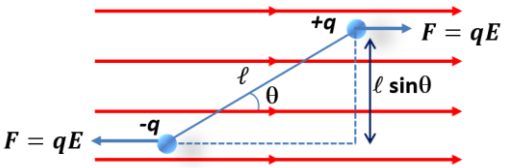
$$= 0 - \frac{1}{2}\left(\frac{qE}{m}\right)\left(\frac{\ell}{v}\right)^2$$

$$|S_y| = \frac{qE\ell^2}{2mv^2}$$

Force and Torque on a dipole in an electric field

Part - 10

Force & Torque on Dipole in a Uniform Electric Field:



In uniform electric field both charges experience equal and opposite force, so net force on dipole is zero.

$F_{net} = 0$

Torque/Couple of force = magnitude of either force × perpendicular distance between lines of force.

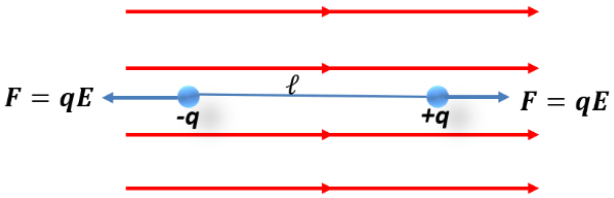
$\vec{\tau} = (qE)l\sin\theta$ or $\vec{\tau} = (ql)E\sin\theta$

$\vec{\tau} = pE\sin\theta$ or $\vec{\tau} = (\vec{p} \times \vec{E})$

When dipole is aligned along the field

Here angle between EF and dipole moment is zero, so

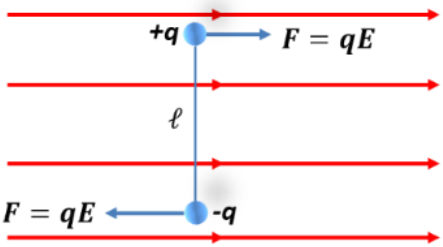
$\tau = pE\sin 0 = 0$ (torque is zero)



When dipole is placed perpendicular to the field.

Here angle between $\vec{E}$ & $\vec{p}$ = 90°, So

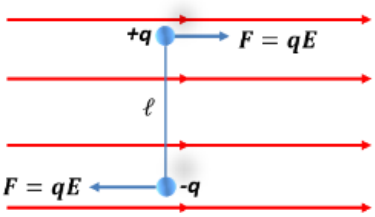
$\Rightarrow \tau = pE\sin 90 = pE(max)$ (Torque on dipole is maximum)



$\tau_{max} = pEsin90^o = PE$

Illustration 1:

If dipole is released from given situation, in which direction it will start rotating.



(1) Clockwise **(2) Anticlockwise**

Solution :

It will start rotating clockwise.

Illustration 2:

When a dipole is placed in a uniform electric field, net force and net torque on it, is always zero. This statement is -

(1) true **(2) false**

Solution :

In uniform electric field net force is always zero but net torque may or may not be zero always, so statement is false.

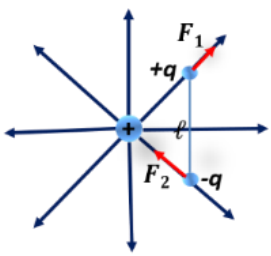
Illustration 3:

When a dipole is placed in a nonuniform electric field, net force and net torque on it is always nonzero. This statement is -

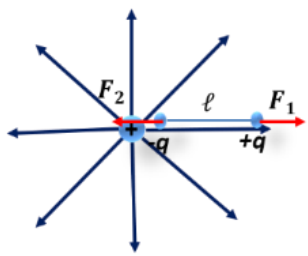
(1) true **(2) false**

Solution :

This statement is false.



Net force and net torque both are non zero

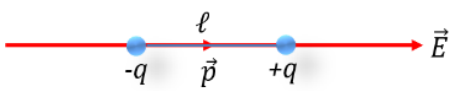


Net force is non zero but net torque is zero

Illustration 4 :

An electric dipole of dipole moment p and moment of inertia I placed in uniform electric field along the field and rotates slightly then released. Find time period of oscillation of dipole.

Solution :



Here $\vec{p}$ and $\vec{E}$ are in same direction so net torque on dipole is zero

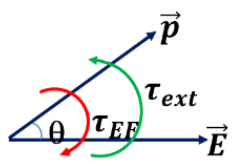
When dipole is rotated in anticlockwise direction slightly and is released, it will experiences a torque in clockwise direction (restoring torque) which is responsible for oscillation of dipole.

Here restoring torque is τ = pEsinθ

$\vec{\tau} = (\vec{p} \times \vec{E})$

As angle is very small so sinθ can be written as θ

$\tau = pE\theta = I\alpha$



From $T = 2\pi\sqrt{\frac{I}{C}} \Rightarrow T = 2\pi\sqrt{\frac{I}{pE}}$ (Here C = pE)

Work done in Rotating a Dipole

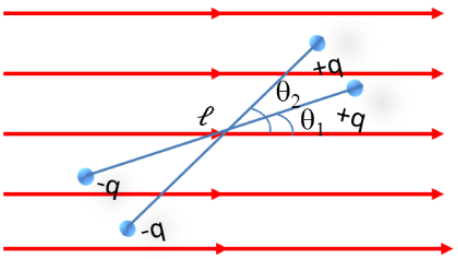
Part - 11

Work done in rotating a dipole in uniform electric field:

Work done in rotating a dipole by small angle dθ is

$dW = \tau d\theta$

Work done in rotating the dipole from initial angular position θ_1 to θ_2 is



$$\int dW = \int_{\theta_1}^{\theta_2} \tau d\theta = pE \int_{\theta_1}^{\theta_2} \sin\theta d\theta \qquad [\because \tau = PE\sin\theta]$$

$$W = pE\left[-\cos\theta\right]_{\theta_1}^{\theta_2} = pE\left(-\cos\theta_2 + \cos\theta_1\right)$$

$$W = pE\left(\cos\theta_1 - \cos\theta_2\right)$$

Illustration 1 :

A dipole of dipole moment 2×10^{-14} Cm is placed in a uniform electric field of $5\times10^4 N/C$. Find work done in rotating the dipole from $60°$ to $90°$.

Solution:

$$W = pE\left(\cos\theta_1 - \cos\theta_2\right)$$

$$W = 2\times10^{-14}\times5\times10^4\left(\cos60° - \cos90°\right)$$

$$W = 5\times10^{-10} J$$

Illustration 2 :

Work done in rotating dipole from 0° to 60° is W. Find torque required to hold the dipole at this position.

Solution:

W = pE (cos0° – cos60°)

$$W = \left(\frac{pE}{2}\right)$$

Torque required to hold the dipole at 60° position

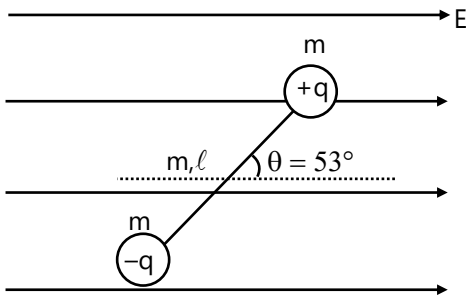
τ = pEsin60°

$$\tau = pE\left(\frac{\sqrt{3}}{2}\right)$$

$$= \sqrt{3}W$$

Illustration 3:

Two-point masses of mass m and equal and opposite charge of magnitude q are attached on the corners of a non-conducting uniform rod of mass m and the system is released from rest in uniform electric field E as shown in the figure from $\theta = 53°$



(i) Find angular acceleration of the rod just after releasing

(ii) What will be angular velocity of the rod when it passes through stable equilibrium.

(iii) Find work required to rotate the system by 180°.

Solution:

(i) $\tau_{net} = PE \sin 53° = I\,\alpha$ $\left(\text{Where } I = \frac{m\ell^2}{12} + m\left(\frac{\ell}{2}\right)^2 + m\left(\frac{\ell}{2}\right)^2 = \frac{7m\ell^2}{12}\right)$

$$\therefore \quad \alpha = \frac{(q\ell)\ E\ \left(\frac{4}{5}\right)}{\frac{m\ell^2}{12} + m\left(\frac{\ell}{2}\right)^2 + m\left(\frac{\ell}{2}\right)^2} = \frac{48qE}{35\,m\ell}$$

(ii) From energy conservation: $K_i + U_i = K_f + U_f$

$$\therefore \quad 0 + (-PE\cos 53°) = \frac{1}{2} I\omega^2 + (-PE\cos 0°)$$

$$\therefore \quad \frac{1}{2} I\omega^2 = PE\,(1 - 3/5) = \frac{2}{5} PE$$

$$\therefore \quad \frac{1}{2} \times \frac{7m\ell^2}{12} \times \omega^2 = \frac{2}{5}\ q\,\ell\,E \qquad \text{or} \quad \omega = \sqrt{\frac{48qE}{35\,m\ell}}$$

(iii) $W_{ext} = U_f - U_i$

$$\therefore \quad W_{ext} = (-PE\cos(180° + 53°)) - (-PE\cos 53°)$$

or $$W_{ext} = (q\,\ell\,)E\left(\frac{3}{5}\right) + (q\,\ell\,)E\left(\frac{3}{5}\right) \qquad \Rightarrow \quad W_{ext} = \left(\frac{6}{5}\right) q\,\ell\,E$$

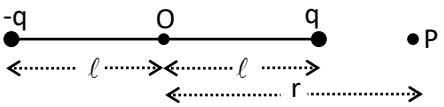
Electrostatic Potential Due to Dipole

Part - 12

Electric Potential due to Dipole:

- **At axial point**

Electric potential due to +q charge $V_1 = \frac{kq}{(r-\ell)}$



Electric potential due to -q charge $V_2 = \frac{-kq}{(r+\ell)}$

Net electric potential $V = V_1 + V_2 = \frac{kq}{(r-\ell)} + \frac{-kq}{(r+\ell)} = \frac{kq \times 2\ell}{(r^2-\ell^2)} = \frac{kp}{r^2-\ell^2}$

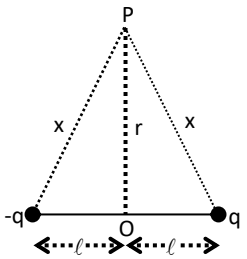
If $r >> \ell$ then $V = \frac{kp}{r^2}$

- **At equatorial point**

Electric potential at P due to +q charge $V_1 = \frac{kq}{x}$

Electric potential at P due to -q charge $V_2 = \frac{-kq}{x}$

Net potential $V = V_1 + V_2 = \frac{kq}{x} - \frac{kq}{x} = 0 \quad (\therefore V = 0)$



- **Potential at general point (r, θ) :** Lets resolve the dipole moment $\vec{P}$ into components : $P\cos\theta$ along radial line and $P\sin\theta \perp$ to the radial line

For the $P\cos\theta$ component, the point A is an axial point,

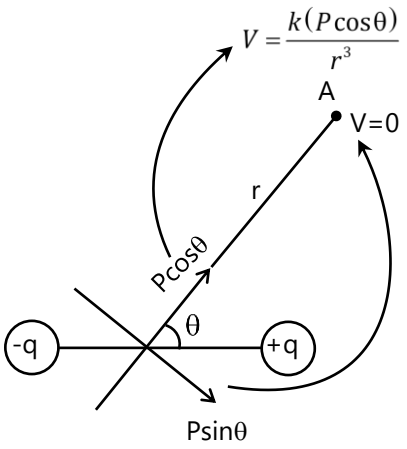
So, potential at A due to $P\cos\theta$ = $\frac{k(P\cos\theta)}{r^2}$

And for $P\sin\theta$ component, the point A is an equatorial point,

So potential at A due to $P\sin\theta = 0$

$$V_{net} = \frac{k(P\cos\theta)}{r^2}$$

$$\therefore \quad V = \frac{k\left(\vec{P}.\vec{r}\right)}{r^3}$$



Electrostatic Potential Energy of Dipole in Electric Field

Part - 13

Potential Energy of dipole in electric field:

$\vec{E}$ is a conservative field so whatever work is done in rotating a dipole from θ_1 to θ_2 it is just equal to change in the electrostatic potential energy.

Work done $W = pE(\cos\theta_1 - \cos\theta_2) = U_2 - U_1$

$\Rightarrow pE\cos\theta_1 - pE\cos\theta_2 = U_2 - U_1$

$\Rightarrow (-pE\cos\theta_2) - (-pE\cos\theta_1) = U_2 - U_1$

$\Rightarrow U_2 = -pE\cos\theta_2$ or $U_1 = -pE\cos\theta_1$

$\Rightarrow U = -pE\cos\theta$ or $U = \vec{p}.\vec{E}$.

Case-(i) : When $\vec{E} \parallel \vec{p} \Rightarrow \theta = 0° \Rightarrow U = -pE = U_{min}$

($\because$ Here F = 0 $\Rightarrow$ ***stable equilibrium****)*

Case-(ii) : When $\vec{E} \perp \vec{p} \Rightarrow \theta = 90° \Rightarrow$ U = 0 *(reference position)*

(Here F = maximum)

Case-(iii) : When angle between $\vec{E}$ & $\vec{p}$ is 180°.

$\Rightarrow U = +pE = U_{max}$ max, but here F = 0

So, called ***unstable equilibrium.***

θ	$\tau = pE\sin\theta$	$U = -pE\cos\theta$	Nature of equilibrium
0°	0	–pE	Stable equilibrium
90°	max	0	Not in equilibrium
180°	0	+pE	Unstable equilibrium

Force on electric dipole in non-uniform electric field :

From relation between force and potential energy

$$F = -\frac{dU}{dr} = \frac{d(p\cos\theta)E}{dr} = p\cos\theta\left(\frac{dE}{dr}\right)$$

$$F = p\cos\theta\left(\frac{dE}{dr}\right)$$

In uniform EF

$\left(\frac{dE}{dr} = 0\right)$

F = 0

$\tau = pE\sin\theta$

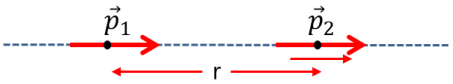
In non-uniform EF

$\left(\frac{dE}{dr} \neq 0\right)$

τ and F may or may not be zero

Dipole-Dipole Interaction:

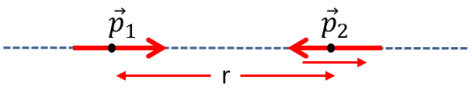
Case-(i) :



Force on p_2 due to p_1 will be

$F_2 = \left(-\frac{6kp_1p_2}{r^4}\right)$ (Negative sign shows force of attraction between both dipoles.)

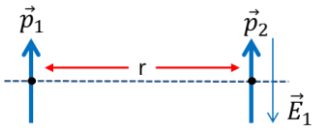
Case-(ii) :



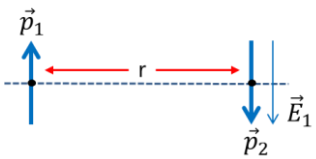
Force on p_2 due to p_1 will be

$F_2 = \left(+\frac{6kp_1p_2}{r^4}\right)$ (Positive sign shows force of repulsion between both dipoles.)

Case-(iii) :



$F = \left(+\frac{3kp_1p_2}{r^4}\right)$

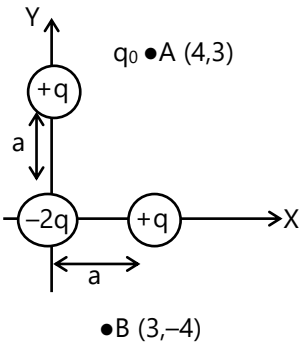


$F = \left(-\frac{3kp_1p_2}{r^4}\right)$

Illustration 1:

(i) Find potential at point A and B due to the small charge system fixed near origin. (Distance between the charges is negligible).

(ii) Find work done to bring a test charge q_0 from point A to point B, slowly. All parameters are in S.I. units.



Solution:

(i) Dipole moment of the system is $\vec{P} = (qa)\,\hat{i} + (qa)\,\hat{j}$

Potential at point A due to the dipole

$$V_A = K\,\frac{(\vec{P}\cdot\vec{r})}{r^3} = \frac{K[(qa)\hat{i}+(qa)\hat{j}]\cdot(4\hat{i}+3\hat{j})}{5^3} = \frac{(7)k(qa)}{125}$$

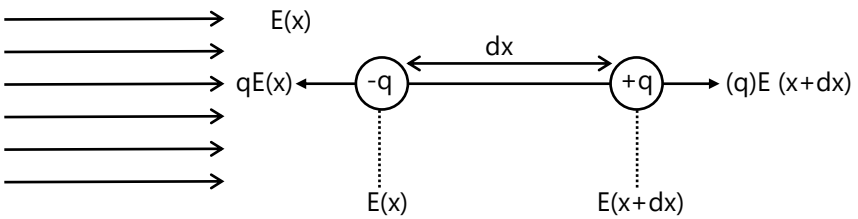
$$V_B = \frac{K\left[(qa)\hat{i}+(qa)\hat{j}\right]\cdot\left(3\hat{i}-4\hat{j}\right)}{(5)^3} = \frac{-K(qa)}{125}$$

(ii) $$W_{A\to B} = U_B - U_A = q_0\,(V_B - V_A) = q_0\left[-\frac{K(qa)}{125} - \left(\frac{K(qa)(7)}{125}\right)\right]$$

$$\Rightarrow \qquad W_{A\to B} = \frac{-Kqq_0a}{125}(8)$$

Dipole in non-uniform electric field:

If the dipole is placed along $\vec{E}$ (shown in figure)



Then, Net force on the dipole : $F_{net} = qE(x + dx) - qE(x)$

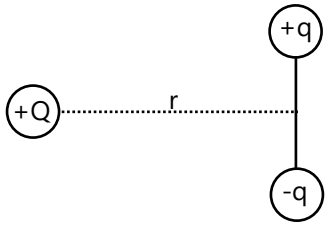
$$F_{net} = q\,\frac{E(x+dx)-E(x)}{dx}\;(dx)\;;$$

Here $q(dx) = P$

$$\therefore\ \mathrm{F_{net}} = \mathrm{P}\left(\frac{\mathrm{dE}}{\mathrm{dx}}\right)$$

Illustration 2:

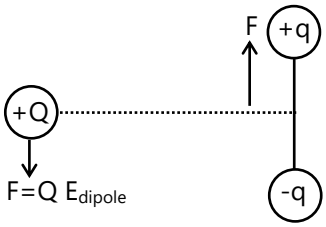
A short dipole of dipole moment P is placed near a point charge Q as shown in figure. Find force on the dipole due to the point charge.



Solution:

Force on the point charge due to dipole $F = (Q)\ (E_{dipole})$

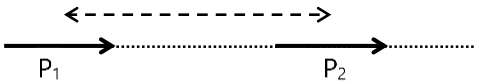
$$F = (Q)\left(\frac{KP}{r^3}\right)(\downarrow)$$



So, force on the dipole due to the point charge will also be $F = \left(\frac{KPQ}{r^3}\right)(\uparrow)$ (but in opposite direction) as shown

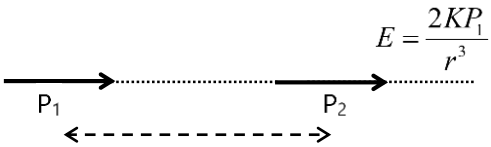
Illustration 3:

Find force on short dipole P_2 due to short dipole P_1 if they are placed at a distance r a part as shown in figure.



Solution:

Force on P_2 due to P_1



$$F_2 = (P_2)\left(\frac{dE_1}{dr}\right)$$

$$\therefore \quad F_2 = (P_2)\left(\frac{d}{dr}\left(\frac{2\ kP_1}{r^3}\right)\right) \text{ or } \quad F_2 = -\frac{6KP_1P_2}{r^4}$$

Here – sign indicates that this force will be attractive (opposite to r).

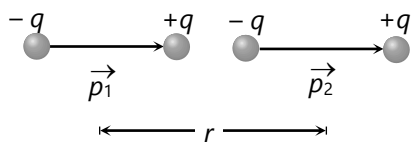
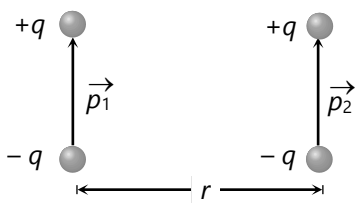
Crux Pan

- For a dipole, potential is zero at equatorial point, while at any finite point E ≠ 0
- In a uniform $\vec{E}$, dipole may experience a torque but may not experience a force.
- If a dipole is placed in a field due a point charge, then torque on dipole may be zero.

Distribution	Point charge	Dipole	Quadrapole	Octapole
V proportional to	r^{-1}	r^{-2}	r^{-3}	r^{-4}
E proportional to	r^{-2}	r^{-3}	r^{-4}	r^{-5}

Force between	Two Point charges	Dipole and point charge	Dipole-dipole
Proportional to	r^{-2}	r^{-3}	r^{-4}

Table: Dipole-dipole interaction

Relative position of dipole	Force	Potential energy
	$\frac{1}{4\pi\varepsilon_0}.\frac{6p_1p_2}{r^4}$ (attractive)	$\frac{1}{4\pi\varepsilon_0}.\frac{2p_1p_2}{r^3}$
	$\frac{1}{4\pi\varepsilon_0}.\frac{3p_1p_2}{r^4}$ (repulsive)	$\frac{1}{4\pi\varepsilon_0}.\frac{p_1p_2}{r^3}$

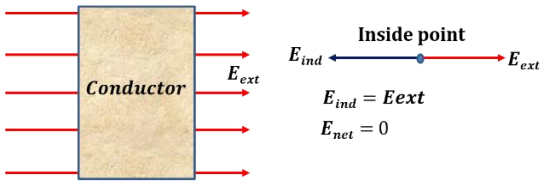
Conductors and their Properties

Part - 14

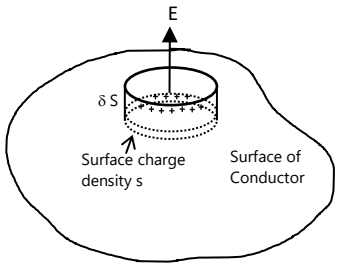
Conductors and their Properties

(i) Conductors are materials which contain large number of free electrons which can move freely inside the conductor.

(ii) Because there is no electric field inside the conductor so we can say electrostatic field lines never exist within conducting materials.



(iii) Electric field is always perpendicular to conducting surface.



(iv) In electrostatics, conductors are always equipotential surfaces.

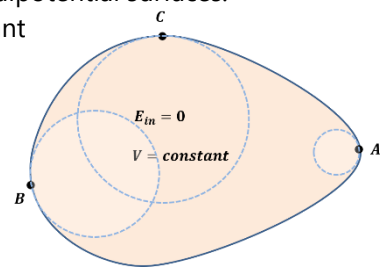
(v) For any conductor at its surface, V= constant

$V_A = V_B = V_C$

$R_A < R_B < R_C$

∵ For conductors σR = constant

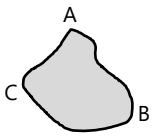
So $\sigma_A > \sigma_B > \sigma_C$



(vi) Charge always resides on the outer surface of a conductor.

(vii) If there is a cavity inside a charged conductor with the cavity devoid of any charge then charge will always reside only on the outer surface of the conductor.

(viii) Electric field intensity near a conducting surface is given by the formula $\vec{E} = \frac{\sigma}{\varepsilon_0}\hat{n}$

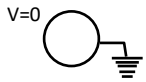


$\vec{E}_A = \frac{\sigma_A}{\varepsilon_0}\hat{n}$; $\vec{E}_B = \frac{\sigma_B}{\varepsilon_0}\hat{n}$; $\vec{E}_C = \frac{\sigma_C}{\varepsilon_0}\hat{n}$

Because conductors are always equipotential surfaces so we can say $V_A = V_B = V_C$

It means $q \propto R$ and $\sigma \propto \frac{1}{R}$ so we can say $\frac{q_1}{q_2} = \frac{R_1}{R_2}$ and $\frac{\sigma_1}{\sigma_2} = \frac{R_2}{R_1}$

(ix) When a conductor is grounded its potential becomes zero.



This concept is known as earthing, after which potential of body becomes zero but charge may or may not be zero. If isolated body is earthed then only we can say that its charge is zero.

Induction in Conductor:

Suppose a neutral conductor is placed in an electric field E. Due to the electric field, many electrons starts shifting in opposite direction to the electric field and get accumulated on the left side. On the right side there will be deficiency of electrons, so positive charge will appear on the right side. These charges are called induced charges and this process is called charge induction. This process continues till the electric field due to the induced charges in the conducting material, balances the external electric field.

Illustration 1 :

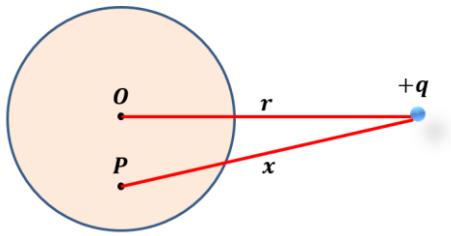
For the below neutral conducting sphere find -

(1) E due to q at P

(2) E due to sphere at P

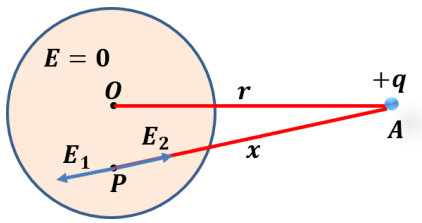
(3) Electric potential at P due to q

(4) Electric potential at P due to sphere



Neutral conducting sphere

Solution :



In every conductor net electric field is always zero, so electric field at P will be equal and opposite due to point charge and induced charges on conductor.

(1) E due to $+q$ charge at P will be $\mathrm{E_1} = \frac{\mathrm{kq}}{\mathrm{x^2}}\left(\text{along}\ \overrightarrow{\mathrm{AP}}\right)$

(2) E due induce charges charge at P will be

$\mathrm{E_2} = \frac{\mathrm{kq}}{\mathrm{x^2}}\left(\text{opposite to}\ \overrightarrow{\mathrm{AP}}\right)$

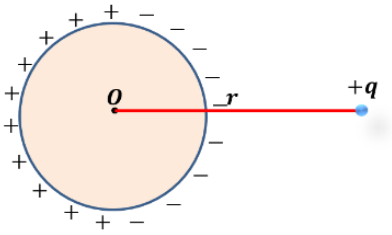
At Point P

$\vec{E} = \vec{E}_1 + \vec{E}_2$

$\vec{E} = 0$

$\vec{E}_1 = -\vec{E}_2$ (i.e. both EF are equal and opposite).

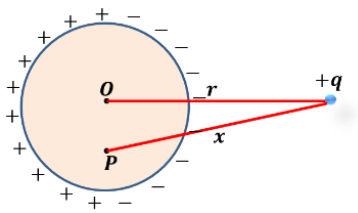
(3)



Potential at centre due to induced charges is zero, and due to +q is $V_q = \frac{kq}{r}$

As potential at every point inside the conductor is same so potential at P is also $V = \frac{kq}{r}$

(4)



At point P

$$V_{ind} + V_q = \frac{kq}{r}$$

$$V_{ind} + \frac{kq}{x} = \frac{kq}{r}$$

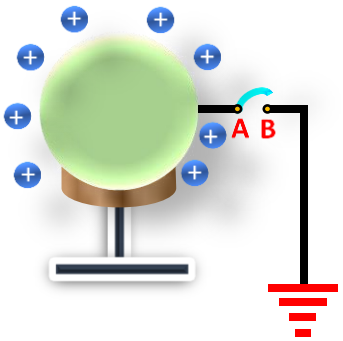
$$V_{ind} = \frac{kq}{r} - \frac{kq}{x}$$

Earthing of Conductor

Part - 15

Concept of Earthing

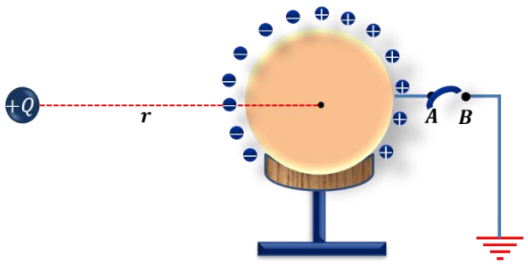
Earth is conducting body with zero potential (assumed). When a charged body is connected to earth, charge flows continuously from higher potential to lower potential till potential of body becomes zero.



As soon as switch is closed whole charge of sphere goes to earth to make potential of sphere zero

Illustration 1 :

What final charge will be appearing on conducting sphere when switch is closed.



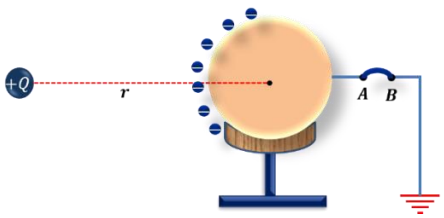
Solution:

If x is charge (which would be appearing) on sphere after closing the switch then,

$$\frac{kQ}{r} + \frac{kx}{R} = 0$$

(Potential due to charge appears on sphere)

(Potential due to point charge)



From $\frac{kQ}{r}+\frac{kx}{R}=0$

$\Rightarrow \frac{kQ}{r}=-\frac{kx}{R}$

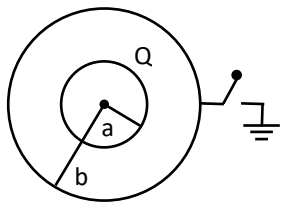
$\Rightarrow x=-Q\left(\frac{R}{r}\right)$

As $R>r$ magnitude of x will be smaller than Q.

When earthing is removed charge on sphere remains at same position due to force of attraction of +Q charge. When point charge +Q is removed whole negative charge of sphere is distributed uniformly on surface of conductor.

Illustration 2 :

Find the charge of outer sphere after closing the switch



Solution:

After closing the switch outer sphere will get earthed so its potential will become zero. Let us assume Q′ charge appears on it finally after earthing -

$\Rightarrow V_{out} = 0$

$\Rightarrow \frac{kQ}{b}+\frac{kQ'}{b}=0$

$\Rightarrow Q' = -Q$

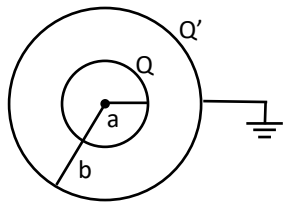
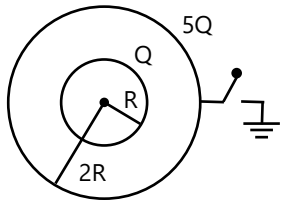


Illustration 3 :

Find the charge transferred from body to earth after closing the switch



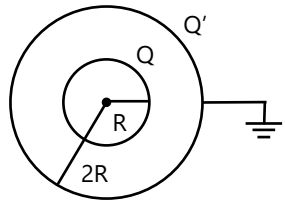
Solution:

After closing the switch outer sphere will get earthed so its potential will become zero, Let us assume Q′ charge appears on it finally after earthing.

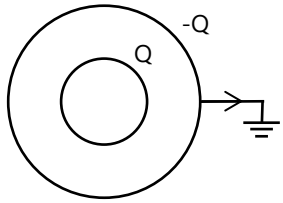
$\Rightarrow V_{out} = 0$

$\Rightarrow \frac{kQ}{2R}+\frac{kQ'}{2R}=0$

$\Rightarrow Q' = -Q$



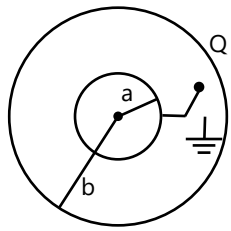
Initially outer sphere was having 5Q charge and finally it becomes -Q that means 6Q charged transferred from body to earth. (or -6Q from earth to body)



$\Rightarrow Q_{Transfer}$ = 6Q from Body to earth -6Q from earth to Body.

Illustration 4 :

Find the charge on inner sphere after closing the switch



Solution:

After closing the switch inner sphere will get earthed so its potential will become zero. Let us assume Q′ charge appears on it finally after earthing.

$\Rightarrow V_{in} = 0$

$\Rightarrow \frac{kQ'}{a} + \frac{kQ}{b} = 0$

$\Rightarrow Q' = -\frac{Qa}{b}$

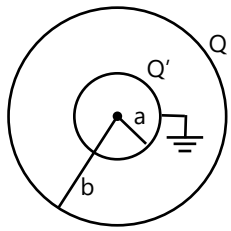
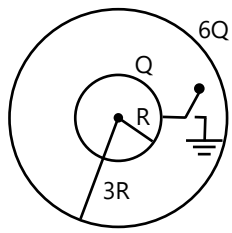


Illustration 5 :

Find the charge transferred from body to earth after closing the switch



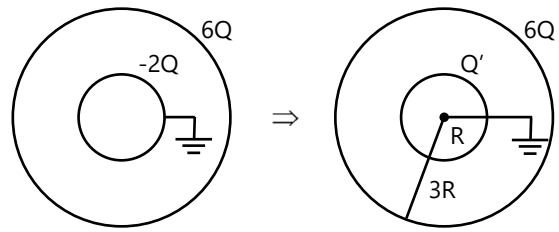
Solution:

After closing the switch inner sphere will get earthed so its potential will become zero. Let us assume Q′ charge appears on it finally after earthing

$\Rightarrow V_{in} = 0$

$\Rightarrow \frac{kQ'}{R} + \frac{k6Q}{3R} = 0$

$\Rightarrow Q' = -2Q$



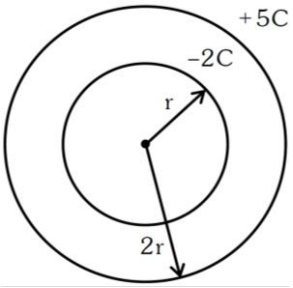
Initially outer sphere was having 6Q charge and finally it becomes -2Q that means 8Q charged transferred from body to earth. (or -8Q from earth to body)

Illustration 6 :

The inner shell is now earthed. What amount of charge will flow from

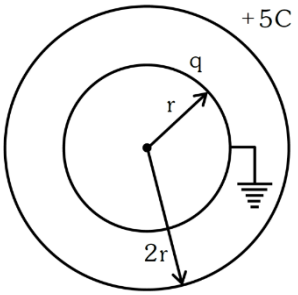
(a) Shell to ground

(b) ground to shell on earthing.



Solution:

(a) If q we the final charge on inner sphere after earthing then

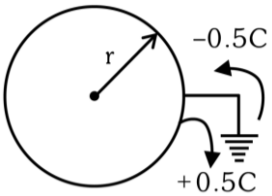


$$q = -Q\frac{r_1}{r_2}$$

$$q = -5\left(\frac{r}{2r}\right) = -2.5C$$

charge that flows from ground to shell $= 2.5 - 2 = -0.5C$

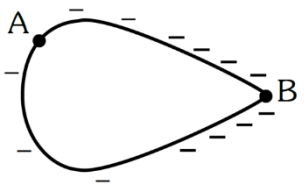
(b) Charge that flows from shell to ground = -2-(-2.5) $= 0.5C$



Leakage of Charge, Electrostatic Shielding and Sharing of Charges

Part - 16

Leakage of Charge



Leakage of charge from sharp points of a conductor

Here $V_A = V_B$ but $\sigma_B > \sigma_A$ & hence $E_B > E_A$

$$\left\{\because E = \frac{\sigma}{\varepsilon_0}\right\}$$

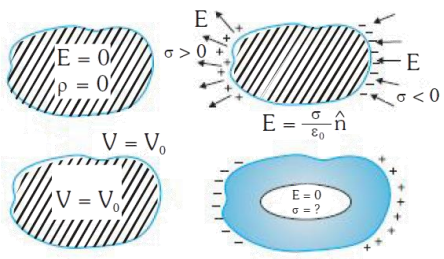
Thus charge density & hence electric field is larger at sharp corners as compared to flat surfaces due to their less radius of curvature.

$$\sigma_1 = \frac{Q_1'}{4\pi R_1^2} \;\&\; \sigma_2 = \frac{Q_2'}{4\pi R_2^2}$$

$$\because \frac{Q_1'}{Q_2'} = \frac{R_1}{R_2} \Rightarrow \frac{\sigma_1}{\sigma_2} = \frac{R_2}{R_1} \Rightarrow \sigma \propto \frac{1}{R}$$

Electrostatic Shielding

- Electrostatic shielding is the method of protecting a certain region from the effect of electric field.
- A cavity surrounded by conducting walls is a field free region as long as there are no charges inside the cavity.
- Whatever be the charge and field configuration the field inside the cavity is always zero (Provided no charge present inside the cavity) This is known as electrostatic shielding.
- Electrostatic shielding can be achieved by enclosing sensitive instruments in a hollow conductor.
- Figure gives a summary of the important electrostatic properties of a conductor.



Sharing of charges

When two charged conductors are connected by a conducting wire then charge flows from a conductor at higher potential to that at lower potential. This flow of charge stops when the potential of both conductors become equal.



A and B both are isolated conducting spheres having initial charges q_1 and q_2. (total charge of system $Q = q_1 + q_2$)

Charge will flow continuously until potential of both spheres becomes equal.

Let the amounts of charges after the conductors are connected be Q_1 and Q_2 respectively and their common potential be V then

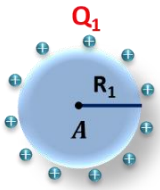


Finally

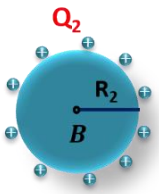
$$V = \frac{kQ_1}{R_1} = \frac{kQ_2}{R_2} \Rightarrow \frac{Q_1}{Q_2} = \frac{R_1}{R_2}$$

or $Q_1 = \left(\frac{R_1}{R_1 + R_2}\right)Q$ or $Q_2 = \left(\frac{R_2}{R_1 + R_2}\right)Q$ $\{Q_1 + Q_2 = Q\}$

Final charge density of spheres



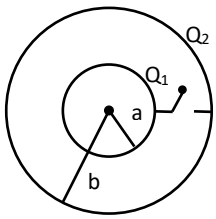
$$\sigma_1 = \frac{Q_1}{A_1} = \frac{\left(\frac{R_1}{R_1 + R_2}\right)Q}{4\pi R_1^2}$$



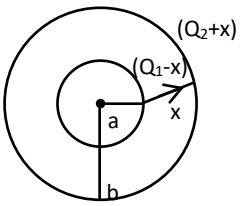
$$\sigma_2 = \frac{Q_2}{A_2} = \frac{\left(\frac{R_2}{R_1 + R_2}\right)Q}{4\pi R_2^2}$$

Illustration 1 :

Find charge on inner sphere after closing the switch



Solution:



$\because$ After closing the switch potential of both the spheres will become equal so $V_{in} = V_{out}$. Now if x charge is flown from inner to outer sphere then equating their potential, we get the below equation

$$\Rightarrow \frac{k(Q_1 - x)}{a} + \frac{k(Q_2 + x)}{b} = \frac{k(Q_1 - x)}{b} + \frac{k(Q_2 + x)}{b}$$

$$\Rightarrow k(Q_1 - x)\left(\frac{1}{a} - \frac{1}{b}\right) = 0$$

$\Rightarrow Q_1 - x = 0$

$\Rightarrow x = Q_1 = Q_{transfer}$ (that means complete charge of inner sphere will get transferred to outer sphere)

Charge on inner sphere = 0 and on outer sphere = $(Q_1 + Q_2)$

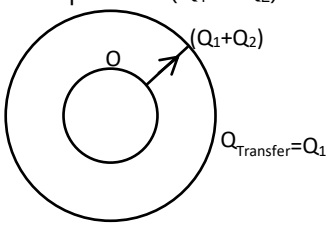
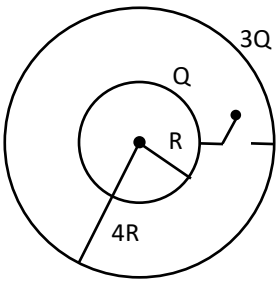


Illustration 2 :

Find the charge of each sphere after closing the switch



Solution:

$\because$ Complete charge of inner sphere will get transferred to outer sphere so Q transfer will be Q

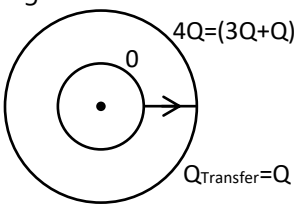
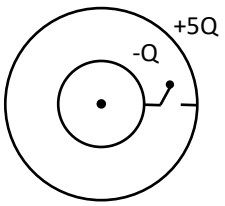


Illustration 3 :

Find the charge transfer after closing the switch.



Solution:

$\because$ Complete charge of inner sphere will get transferred to outer sphere so $Q_{transfer}$ will be -Q

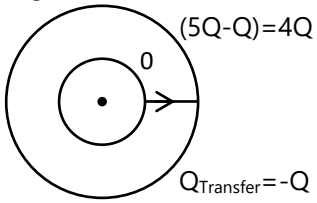


Illustration 4 :

Two thin spherical shells made of metal are at a large distance apart. One, of radius 10 cm, carries a charge of +0.5 mC and the other, of radius 20 cm, carries a charge of +0.7 mC. The charge on each, when they are connected by a suitable conducting wire is respectively -

Solution:

$$Q_1 = \left(\frac{R_1}{R_1 + R_2}\right)Q = \left(\frac{10cm}{30cm}\right)1.2mc$$

$$\Rightarrow Q_1 = 0.4mc$$

$$Q_2 = \left(\frac{R_2}{R_1 + R_2}\right)Q = \left(\frac{20cm}{30cm}\right)1.2mc$$

$$\Rightarrow Q_2 = 0.8mc$$

Illustration 5 :

Two isolated metallic spheres of radii R and 2R are charged such that both of these have same charge density σ. These spheres are located far away from each other and connected by a thin conducting wire, the new charge density on the bigger sphere -

Solution:

$q_1 = \sigma \times 4\pi R^2$ and $q_2 = \sigma \times 4\pi(2R)^2 = \sigma \times 16\pi R^2$

Total charge of system is $Q = q_1 + q_2$

$$Q = \sigma \times 4\pi R^2 + \sigma \times 16\pi R^2 = 20\sigma\pi R^2$$

Final charge on spheres

$$Q_1 = \left(\frac{R}{R + 2R}\right)Q = \left(\frac{1}{3}\right)Q$$

$$\Rightarrow Q_1 = \left(\frac{20\sigma\pi R^2}{3}\right)$$

$$\Rightarrow \sigma_1 = \frac{Q_1}{4\pi R^2} = \left(\frac{20\sigma\pi R^2}{3(4\pi R^2)}\right)$$

$$\Rightarrow \sigma_1 = \left(\frac{5\sigma}{3}\right)$$

Final charge on sphere B

$$\Rightarrow Q_2 = \left(\frac{2R}{R + 2R}\right)Q = \left(\frac{2}{3}\right)Q$$

$$\Rightarrow Q_2 = \left(\frac{40\sigma\pi R^2}{3}\right)$$

$$\Rightarrow \sigma_2 = \frac{Q_2}{4\pi 4R^2} = \left(\frac{40\sigma\pi R^2}{3 \times 16\pi R^2}\right)$$

$$\Rightarrow \sigma_2 = \left(\frac{5\sigma}{6}\right)$$

Illustration 6 :

Two spherical conductors of radii 4 m and 5m, are charged to the same potential. If σ_1 and σ_2 be the respective values of the surface charge density on the two conductors, then the ratio $\frac{\sigma_1}{\sigma_2}$.

Solution:

$$\text{As } \sigma \propto \frac{Q}{R^2}$$

$$\text{so } \frac{\sigma_1}{\sigma_2} = \frac{Q_1}{R_1^2} \times \frac{R_2^2}{Q_2}$$

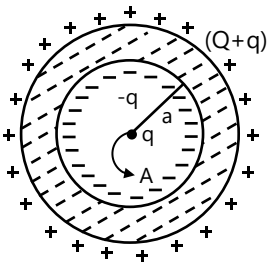
For spherical conductors

$$V = \sigma R$$

$$\text{so } \sigma \propto \frac{1}{R}$$

$$\frac{\sigma_1}{\sigma_2} = \frac{R_2}{R_1} = \frac{5}{4}$$

Cavity inside conductor



Only inner charge will be responsible for charge distribution over inner surface.

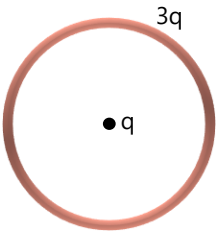
Only outer charge will be responsible for charge distribution over outer surface.

Surface charge density over inner surface $\sigma_{inner} = \frac{-q}{4\pi a^2}$ and outer surface $\sigma_{outer} = \frac{Q+q}{4\pi b^2}$.

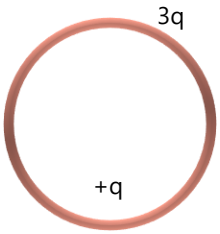
If charge q is shifted to A then charge over inner surface will get distributed non-uniformly and charge distribution over outer surface will be same as before (It would not get effected by inner charge).

Illustration 7 :

Find charge distribution on each surface.

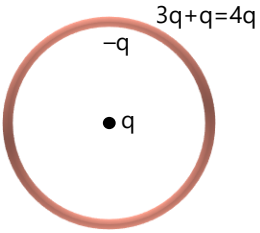


(A)

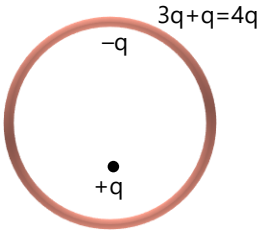


(B)

Solution:



(A)



(B)

Inner $\rightarrow$ −q (uniform distribution)

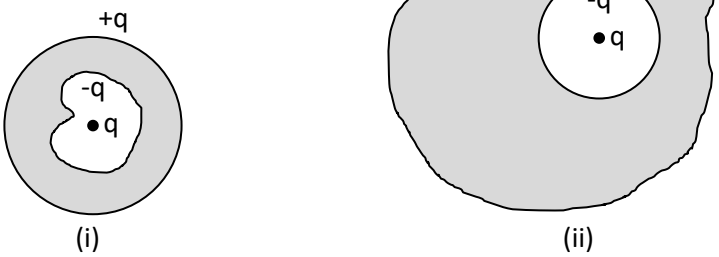
Outer $\rightarrow$ 4q (uniform distribution)

Inner $\rightarrow$ −q (nonuniform distribution)

Outer $\rightarrow$ 4q (uniform distribution)

Illustration 8 :

Find charge distribution on each surface.



Solution:

(i) Inner → –q (Non-uniform distribution)

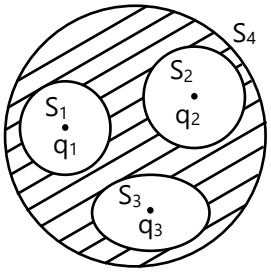
Outer → q (Uniform distribution)

(ii) Inner → –q (Uniform distribution)

Outer → q (Non-uniform distribution)

Illustration 9 :

Find charge distribution on each surface of the body.



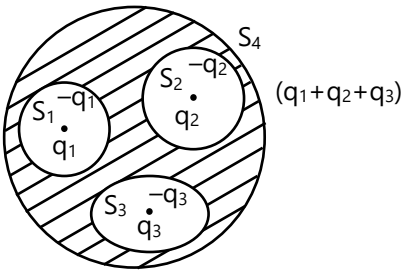
Solution:

$S_1 \rightarrow -q_1$ (Uniform)

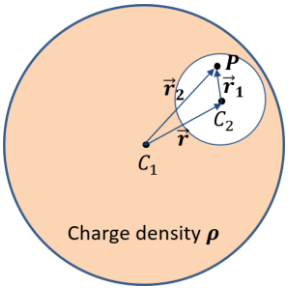
$S_2 \rightarrow -q_2$ (Non-uniform)

$S_3 \rightarrow -q_3$ (Non-uniform)

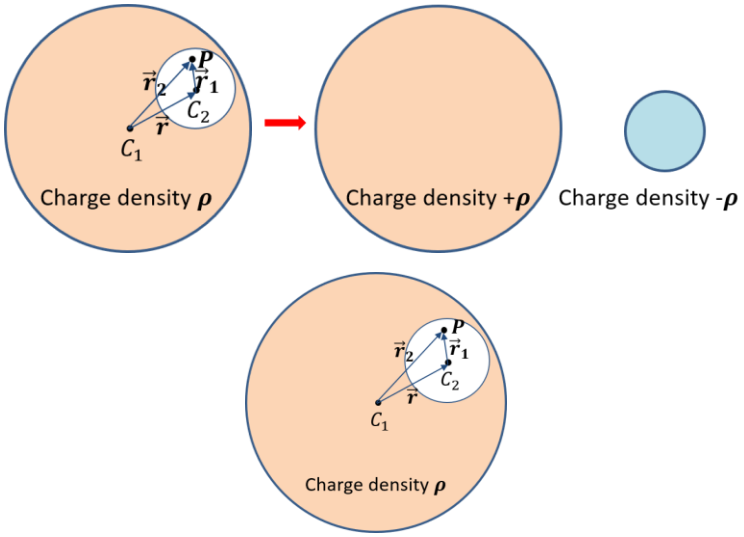
$S_4 \rightarrow q_1+q_2+q_3$ (Uniform)



Electric field in a cavity of uniformly charge solid sphere



For finding electric field intensity at point P in cavity we need to assume small spheres of $+\rho$ and $-\rho$



Electric field at P due to large sphere

$$\vec{E}_1 = \frac{\rho \vec{r}_2}{3\epsilon_0}$$

Electric field at P due to small sphere

$$\vec{E}_2 = \frac{\rho \vec{r}_1}{3\epsilon_0}$$

Net electric field at P

$$\vec{E} = \vec{E}_1 + \vec{E}_2$$

$$\vec{E} = \frac{\rho \vec{r}_2}{3\epsilon_0} - \frac{\rho \vec{r}_1}{3\epsilon_0} = \frac{\rho}{3\epsilon_0}\left(\vec{r}_2 - \vec{r}_1\right) = \frac{\rho}{3\epsilon_0}\vec{r}$$

Hence, electric field is uniform inside cavity.